Predictive HR Analytics: An Interpretable MARS-Machine Learning Pipeline

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Table of Contents

1 Introduction

1.1 Problem Definition

Enterprise turnover costs have long hampered organizations in translating their human capital into profitability and performance. This is more apparent the higher up the value chain an employee is situated or the more specialized an occupation the employee works in. Industry organizations like SHRM which aver that the total **replacement costs can often range between 50 and 200 % of the annual salary** of the departing employee [1].

The deleterious effects of vacated critical high value roles transcend diminished output in that they also materialize as bottlenecks that place cumbersome workloads on the remaining staff, this is no small part owing to information or knowledge siloing in specialty roles [2].

Traditional HR practices in turnover mitigation have been limited to ex post facto analyses (e.g., exit interviews) that amount to forensic examinations of departure with minimal interventional opportunity. These approaches often frame turnover events as lone points of catastrophic failure and in doing so neglect to account for the attenuative nature of turnover.

For instance, in a 2025 Survey, McKinsey reported that 28-36% of all workers report experiencing burnout [3]. When considered with Gallup's findings in 2017 that only 3 in 10 employees believe their opinions are meaningfully considered by the organizations employing them [4], these trends underscore the necessity of HCM and HR Data infrastructure that is robust in being able to proactively flag attrition flight risks before the turnover event occurs.

1.2 Context and Background

Among the ever-present and scantly discussed challenges of the application of Predictive modeling and machine learning in the HR space lies the efficacy of the model architect(s) in disseminating abstract mathematical results from modelling (like model coefficients and gini importance) to HR managers in a format intuitive and intelligible to them.

To bridge this ravine, we will turn to McShane & Von Glinlow's MARS Model of Individual Behavior and channel it's application to feature categorization for a more robust interpretation of model results [5]. The MARS model comes to us from the field of Organizational Behavior and avers that employee behavior can be explained by four pillars:

- **Motivation (M)** : Intrinsic factors that drive employee performance (like agreeableness, initiative etc)
- **Ability (A)**: These represent both natural abilities and acquired skills of employees
- **Role Perception (R)**: These factors represent an employee's understanding of their role in the organization and what is expected of them.

- **Situational Factors (S)**: These factors encompass all conditions that are external to the locus of control of the employee such as time, legal and regulatory environment, macroeconomic conditions etc.

Furthermore, the success of past studies in demonstrating that these pillars directly govern employee behavior merits the exploration of the MARS framework as an interpretive lens in this study [6].

Foundational literature in the realm of Statistical modeling had established a consensus around the theory that drive for high predictive accuracy must supersede interpretability [7]. This is because at the time, algorithmic modeling was seen as an amelioration of the rigid and inflexible stochastic models that preceded them. This theory however has been increasingly nuanced by recent scholarship where concepts like *"Rashomon Sets"* have been employed to argue that for every dataset where there exists a finite number of predictive models, there must exist one interpretable model among them [8]. This lends credence to the viability of interpretable models in producing accurate results that are statistically no different than black box models.

Black box approaches also fail at attributing and locating causal links between attrition factors and turnover events which have been brought to light by scholars like Tambe et al. as looming legal and compliance risks for Enterprise HR teams [9]. Therefore this Capstone study will seek to reject complex ensemble methods in favor of inherently interpretable models like Logistic Regression and Decision Trees to generate actionable odds ratios and logical rules rather than opaque flight-risk scores.

1.3 Objectives and Goals

The choice of *model evaluation metric* is often one of the most pivotal decisions undertaken by a model architect and must pay fidelity to domain or industry specific realities of the dataset at hand. Consequently, In Enterprise HR settings, `Recall` is often favored as metric of choice in Attrition Classification studies as the **false negatives** recall represents are lost turnover intervention opportunities where as **false positives** from `precision` usually amount to a surplus of retention "check-in" conversations with employees that can be more easily absorbed by the enterprise. We will therefore implement `Recall` as one of the primary metrics driving model selection and understanding.

`F1 scores` will complement the use of `Recall` as an evaluative metric in this model as `precision` and `recall` mathematically must involve a tradeoff and `F1 scores` offer a balanced look at the model's overall performance that exceed the value of simple `accuracy` scores which can be inflated by class imbalanced datasets.

Given that the ultimate and foremost objective of this Capstone Study is the construction of a turnover classification modeling pipeline that surpasses binary `yes/no` label predictions with actionable insights like the aforementioned *odds ratios* and *decision rules*, the outputs emanating from Models constructed in this study will be mapped to the **MARS framework** in the hope that enterprise stakeholders can design targeted retention interventions.

1.4 Summary of Approach

To fashion the interpretable attrition classification models requisite to this study, at first standard data cleaning practices will be applied to the datasets employed by this study. After this a benchmark `logistic regression` model is trained on the data with no tuning measures performed to serve as a true baseline. The benchmark will be contrasted with `Regularized Logistic Regression`, `Linear Support Vector Machine` (or `Linear SVM`), `k Nearest Neighbors` classifier (`kNN`), `Decision Trees` and `Generalized Additive Models` (or `GAM's`). Post hoc and *diagnostic tests* will be conducted for each model followed by a consolidated summary or tabulation for model performance which will include an extrapolation of model *feature importance* to **MARS** criteria outlined above to suture the mathematical insights produced to HR context and understanding.

2 Methods

2.1 Data Architecture and Scaling Strategy

The MARS model infused ML pipeline established in the prior section of this capstone will assessed using 3 datasets as part of a broader horizontal scaling strategy to substantiate its real word applications. Therefore the datasets used herein will be the IBM dataset[10] which will be employed as a baseline for methodological and code calibration, the Edward Edward Babushkin Employee Turnover dataset [11] to grapple with feature sparsity and finally, the Data Scientist Job Change dataset[12] intended as a computational test set due to its comparatively larger size.

Below is a more comprehensive overview of the provenance and features of each dataset:

1. IBM dataset (1470 rows, 35 columns): Introduced earlier as the baseline dataset for this study in debugging the pipeline and conducting feature extraction. The IBM HR Attrition Dataset was compiled by Data Scientists at IBM in an effort to synthesize realistic attrition scenario for predictive modelling studies [10]. This dataset has been pruned to 15 features corresponding to the four prongs of the MARS model with non-informative and zero variance columns being prioritized for removal. Below is a summary of feature mappings to their ascribed MARS factors:

- **Motivation (M):** To operationalize the Motivation factor within the IBM dataset, the pipeline extracts the `JobSatisfaction`, `TrainingTimesLastYear` and `StockOptionLevel` Features. `JobSatisfaction` serves as a direct psychological proxy for an employee's internal drive and engagement levels which merits its assignment to the Motivation label.The inclusion of `TrainingTimesLastYear` *(Amount of time spent training in the previous year)* and `StockOptionLevel` *(flag for senior level organizational rewards)* in this category further nuance the exploration of this category by serving as stand-in's for intrinsic and extrinsic motivational factors respectively.
- **Ability (A):** Ability is mathematically encoded through the `PerformanceRating`, `Education`, `EducationField` and `YearsAtCompany` variables. Given that Ability is understood in OB (Organizational Behavior) Research to be a combination of natural aptitude and learned skills. `PerformanceRating` and `YearsAtCompany` will serve here as estimations of learned skill while `Education` and `EducationField` will serve as generalizations of aptitude.
- **Role Perception (R):** To capture this factor the model relies on the `Department`, `JobLevel`, `YearsInCurrentRole` and `YearsWithCurrManager` features. Here `Department` and `JobLevel` dictate the structural clarity of the employee's responsibilities and hierarchical expectations within the enterprise while `YearsInCurrentRole` and `YearsWithCurrManager` account for variance in this factor on account of tenure and longevity.

- **Situational Factors (S):** given their capacity to measure external logistical stressors that impact turnover independent of an employee's actual ability or motivation, the `DistanceFromHome`, `BusinessTravel` and `EnvironmentSatisfaction` columns will be retained in our feature selection phase to model situational constraints

2. Edward Babushkin Employee Turnover dataset (1129 rows, 16 columns): As real world enterprise data operations present challenges to data mining on employees in the form of sparse and un-curated datasets as they're often compiled together with input from multiple departments (HR, Marketing, Payroll, Legal etc.), the Babushkin dataset which is derived from Edward Babushkin, an enterprise HR researcher [11], will serve as a test case in encountering and regulating these data idiosyncrasies.

As Babushkin has sourced this dataset from a native Russian HRIS system, the variables suffer from a heavy "lost-in-translation" effect when ingested by a culturally dissimilar audience.

For example, the target attrition flag column of this dataset is called `event` and is binary encoded.

Other Emic features of this dataset include `stag` which stands for *Stazh* ( ), which is Russian for "Length of Service", it represents tenure in months and will map to the **Ability** factor of the MARS paradigm.

Situational factors in this dataset will be represented by the `greywage`, `way` and `traffic` columns. `way` refers to the commute method of the employee (labels include `bus`, `car` and `foot`), `traffic` is the source of hire ranging from local job aggregators (like the label `youjs`) or referrals (labels `referal` and `recNErab` to represent internal and external referrals respectively) and finally the `greywage` describes the compensation method as it pertains to Russian corporate culture. For this feature, a "white wage" refers to an employee who's salary is 100% official and reported to the tax authorities, In contrast, a "grey wage" means the company pays the minimum wage officially but otherwise remunerates the employee in a circumspect and extra-legal manner.

Role Perception within this sparse architecture is operationalized through the `industry` and `profession` variables, which establish the structural boundaries of the employee's duties. Furthermore, this category uniquely incorporates the `coach` binary feature, serving as a critical proxy for how effectively the enterprise clarified those job expectations via formal mentorship.

Lastly, Babushkin's dataset presents this study with all five factors of the Five factor model (also known as "FFM Model", the most widely accepted and empirically validated framework in personality psychology for describing human personality). Elements of the five factor model and their nomenclature in the Russian HRIS ecosystem as employed by Babushkin are as follows: Extraversion (`extraversion`), Neuroticism (`anxiety`), Conscientiousness (`selfcontrol`), Openness to Experience (`novator`) and Agreeableness (represented on a reverse scale by the `independ` feature as high independence implies low agreeableness). These five columns in tandem will be mapped to the **Motivation** prong of the MARS model and offer us a direct approach in measuring motivation when contrasted against the proxy variables or features that will be employed in other datasets.

3. Data Scientist Job Change dataset (19158 rows, 14 columns): Aggregated by an HR management firm to identify flight risks and poaching vulnerabilities specifically within the hyper-competitive Data Science and tech sector [12], this dataset will be employed to demonstrate the scalability of the MARS-Machine Learning Pipeline to larger datasets which are also often challenged by imperfections like missing data.

The selection of this dataset also exposes the attrition modeling pipeline being constructed in this report to the particularities of bourgeoning and novel field of practice (Data Science) and thereby will allow for an examination of its generalizability to future data via its exposure to recent, sector-specific attrition patterns. This approach ensures the pipeline remains relevant as modern work behaviors evolve.

Contained in this dataset is the attrition column of interest titled `target` and is a binary flag of turnover status. The **Motivation** factor of the MARS model will operationalized by the `enrolled_university` and `last_new_job` features as the university enrollment status of an employee is a direct proxy for their intrinsic motivation to upskill in the form of continuing education. The `last_new_job` feature can be viewed in this context as a quantifiable proxy for career velocity and job-hopping propensity, measuring the temporal gap between an employee's organizational transitions.

The **Ability** prong of MARS will be captured mathematically by the `relevent_experience` and total `experience` (years) columns as in the hyper-specialized tech sector, the competency of data science leadership is optimally elucidated by longitudinal experience, reflecting an accrued mastery of the Enterprise Tech Stack integral to their role.

Role Perception will be represented by the `company_size`, `company_type`, and `training_hours` features. The selection of this subset is contingent on the logic that size and type of the startup/enterprise dictate the structural boundaries of the data scientist's role, while `training_hours` quantifies how much formal onboarding they received to clarify those expectations.

Finally **Situational** factors will be accounted for via the `city_development_index` variable which is fashioned by its authors as a macro-environmental feature that measures the quality of life and economic scaling of the city the employee lives in, serving as an external logistical stressor independent of the employee's actual job performance.

Data Pre-processing Summary: The heterogeneous origins of the datasets employed by this study necessitate standardization through a strict, uniform preprocessing pipeline before being fed into Python's scikit-learn algorithms which will underpin the computational engine of this MARS-Machine Learning pipeline. Given the unintelligibility of plain text strings like `industry` or `commute method` to machine learning models, One-Hot Encoding will be performed to reorient these features to binary sparse matrices (0s and 1s). The ubiquity of magnitude disparities encountered in numerical columns will further compel the use of a scaling strategy like MinMax Scaling which normalize numerical features so that algorithms reliant on spatial geometry and Euclidean distance (like Linear SVM and kNN) are not mathematically distorted by arbitrarily large values. To meet the challenge of missing categorical data (as encountered in the Data Science Job Change dataset), Constant Imputation will be utilized as this study's imputation strategy of choice given that record elimination would result in an infeasible training or sample set size and that modal imputation (imputing each column with the most frequently occurring label) would mathematically distort the reality of the data. The Constant Imputation method stipulates that missing values are uniformly filled with a new standalone string class called `Unknown`. Doing so will allow algorithms availed in this study to evaluate if the sheer lack of enterprise data in itself is actually a predictive behavioral signal for attrition.

2.2 Mathematical Frameworks

The properly establish statistical validity and pattern generalizability across the disparate datasets used in this study and in keeping with Tambe's framework for modern HR analytics, the predictive architecture constructed there in will be underpinned by a comparative panel of six distinct classification algorithms: Standard Logistic Regression, Regularized Logistic Regression (featuring an L2 or Ridge Penalty), Linear Kernel SVM, Decision Tree Classifier, the k-Nearest Neighbors Classifier algorithm and a Generalized Additive Model (GAMs)

1. **Logit Function (Standard Logistic Regression - Model A):** Given that recent studies conducted by scholars like Wu & Fukui [13] have exposed the unsuitability of machine learning operations on HR data with uncalibrated hyperparameters, Standard Logistic Regression will function as a baseline model estimator to observe the amerliorative effects of both regularization (Logistic Regression with L2 penalty) and non-linear pattern identification (Decision Trees).

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0+\beta_1 x_1+\ldots+\beta_n x_n)}}$$

The formula above uses Euler's number (e) to map the linear combination of features into a strict probability curve between 0 and 1and in doing so assumes a linear relationship between predictive features and turnover odds without correction via penalties.

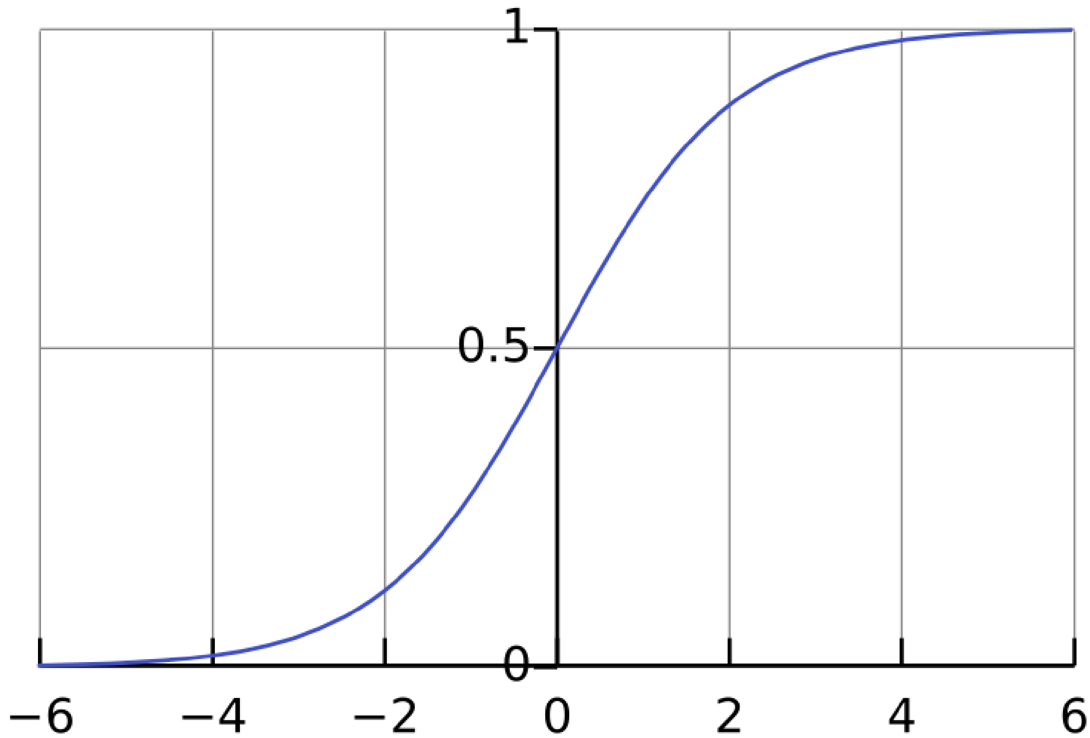


Figure 1: Sigmoid Function Curve

2. **Penalized Loss Function (Ridge/L2 Regularized Logistic Regression - Model B):** To mitigate the deleterious effects of multicollinearity inherent in behavioral HR data and documented by Wu & Fukui in their analysis [13] on the attrition rates in the mental health industry [cite: 1][cite_start], an L2 penalty term is integrated into the Standard logistic regression algorithm to mathematically shrink the noise from overlapping variables (like tenure and age). [cite_start]Save for this remedial step the interactions between collinear features would otherwise come to dominate or compromise the model.

$$J(\theta) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(h_\theta(x_i)) + (1-y_i)\log(1-h_\theta(x_i))] + \lambda\sum_{j=1}^{p}\theta_j^2$$

The loss function $J(\theta)$ formulated above thereby includes a strict penalty term (λ) that penalizes coefficient magnitude, forcing the algorithm to constrain variance and in explicit terms produce a more stable model than the unregulated benchmark.

3. **Linear Kernel Support Vector Machines (Linear SVM - Model C):** Support vector machine algorithms seek to construct a decision boundary by solving a constrained

optimization problem. The selection of this technique enriches the overall methodology of this study with its contribution to the algorithmic diversity of the comparative panel of models being developed in this section. This is attributable to SVM's abandonment of probabilities (like those derived from Logistic Regression models) for geometry in favor of building a physical decision boundary (a hyperplane) to separate those who quit from those who stay. Although more sophisticated kernels such as the Radial Basis Function (RBF) are implementable to Support Vector Machines, confining our SVM to the Linear Kernel allows us to pay fidelity to this Study's core objective of interpretability [8].

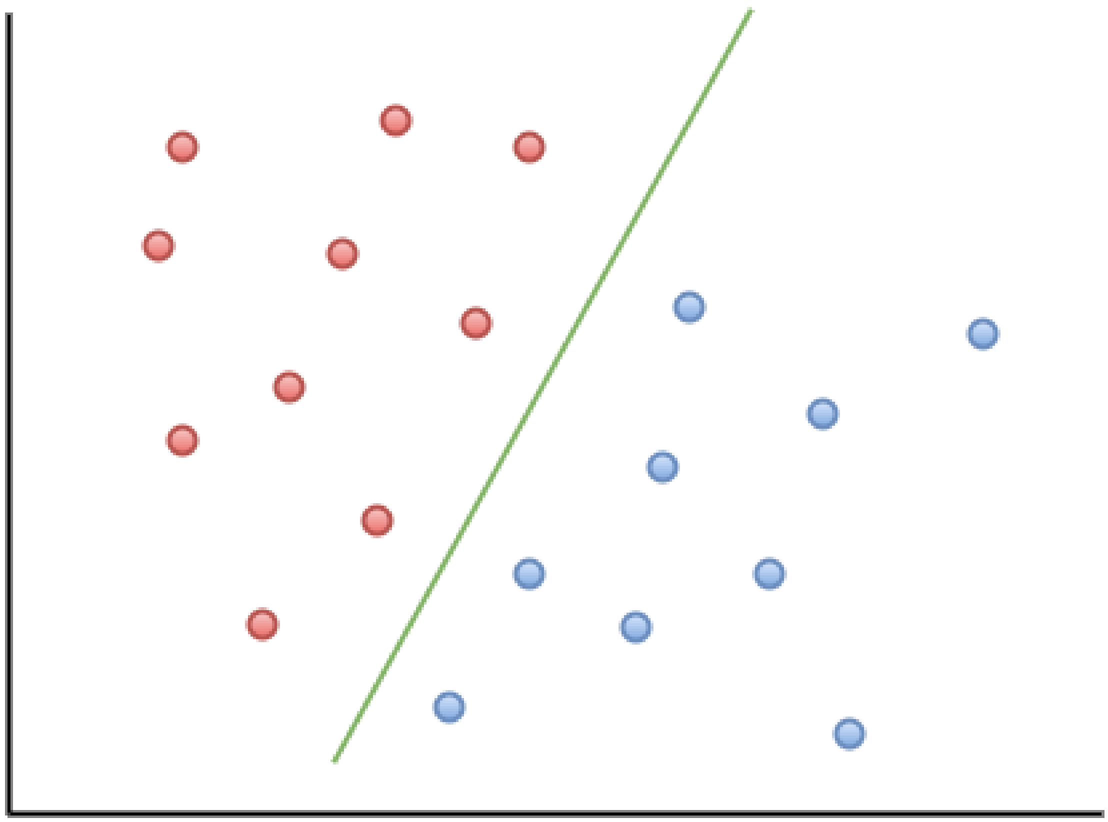


Figure 2: SVM Soft Margin Hyperplane

$$\min_{w,b,\xi} \frac{1}{2}||w||^2 + C\sum_{i=1}^{n}\xi_i$$

$$\text{Subject to: } y_i(w \cdot x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

The formula above attempts to mediate between two contradictory goals. The first half ($||w||^2$) seeks to maximize the width of the hyperplane separating stayers and leavers. The latter half

$(C \sum \xi_i)$ serves as a penalty term for each mistake the model makes. Finally, the slack variable (ξ_i) functions as a bulwark against model rigidity and overfitting by permitting the model to misclassify a limited number of outliers, provided that the interpretability and straightness of the decision boundary are maintained.

4. **Decision Trees (Model D):** To properly account for and accommodate non linear variability in the predictive ensemble of models being assembled, Decision Trees will be implemented to adapt to the irregularities overlooked by linear models like Logistic Regression, Linear SVM and kNN.

As decision trees are computed by splitting datasets on hierachichal and branching conditional logic which materializes in a tree shaped diagram, one of two evaluative loss functions seen below govern the distribution of individual records to individual nodes at the bottom level known as "leaves" [14].

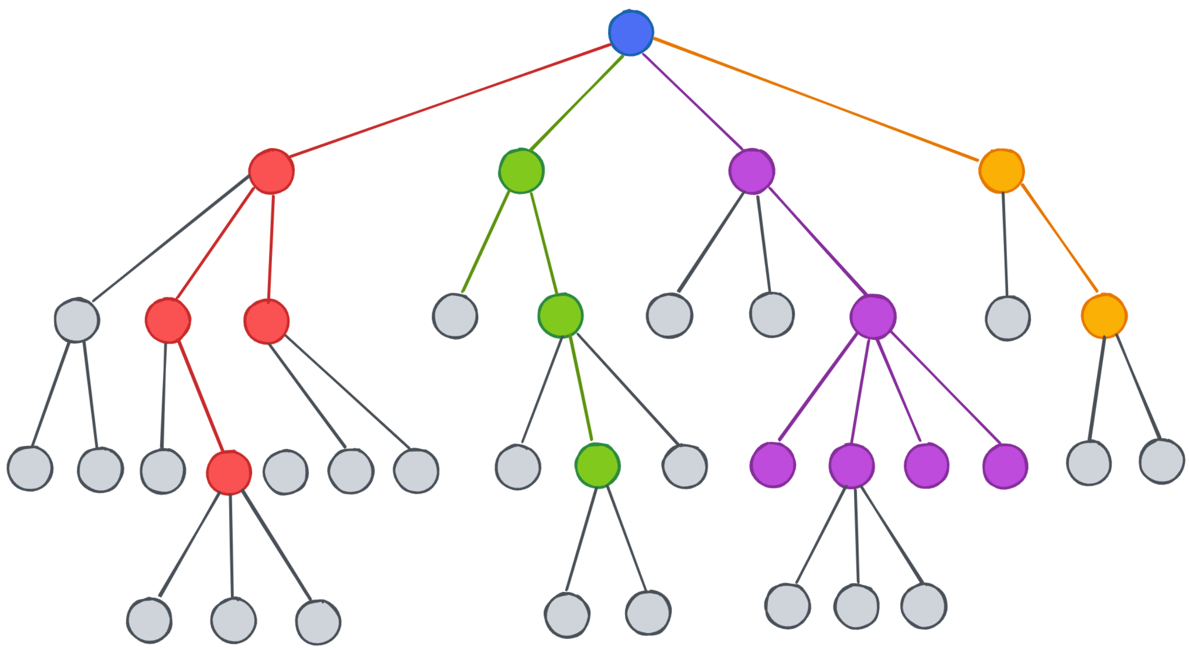


Figure 3: Decision Tree Classifier

$$Gini(p) = 1 - \sum_{i=1}^{C} p_i^2$$

Gini Impurity formulated above measures the probability of incorrectly classifying an employee. Decision trees (specifically CART) aim to minimize this value, aiming for a Gini score of 0, meaning all data points in a leaf node belong to a single class as during tree construction all permutations of splits are evaluated in converging lowest weighted Gini impurity for the child nodes.

$$H(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)$$

Entropy however, approaches Tree optimization by quantifying the uncertainty of a target variable, representing how intermingled the classes are in a node or "leaf". High entropy (E = 1.0) is observed when the node or has an even split of classes (e.g., 50% attrition, 50% retention), conversely low entropy (E = 0) is achieved with maximal node purity.

5. **Euclidean Distance (k-Nearest Neighbhors (kNN) Classifier - Model E):** kNN Classifiers merit selection into the MARS-Machine Learning pipeline owing to their sweeping abandonment of model coefficients in favor of spatial geometry, by obtaining the literal squared mathematical distance ($d(x, y)$) across all dimensions between a target employee and their closest neighbors to predict turnover based on peer resemblance which translates to its unique suitability in disseminating results to audiences in an HR context.

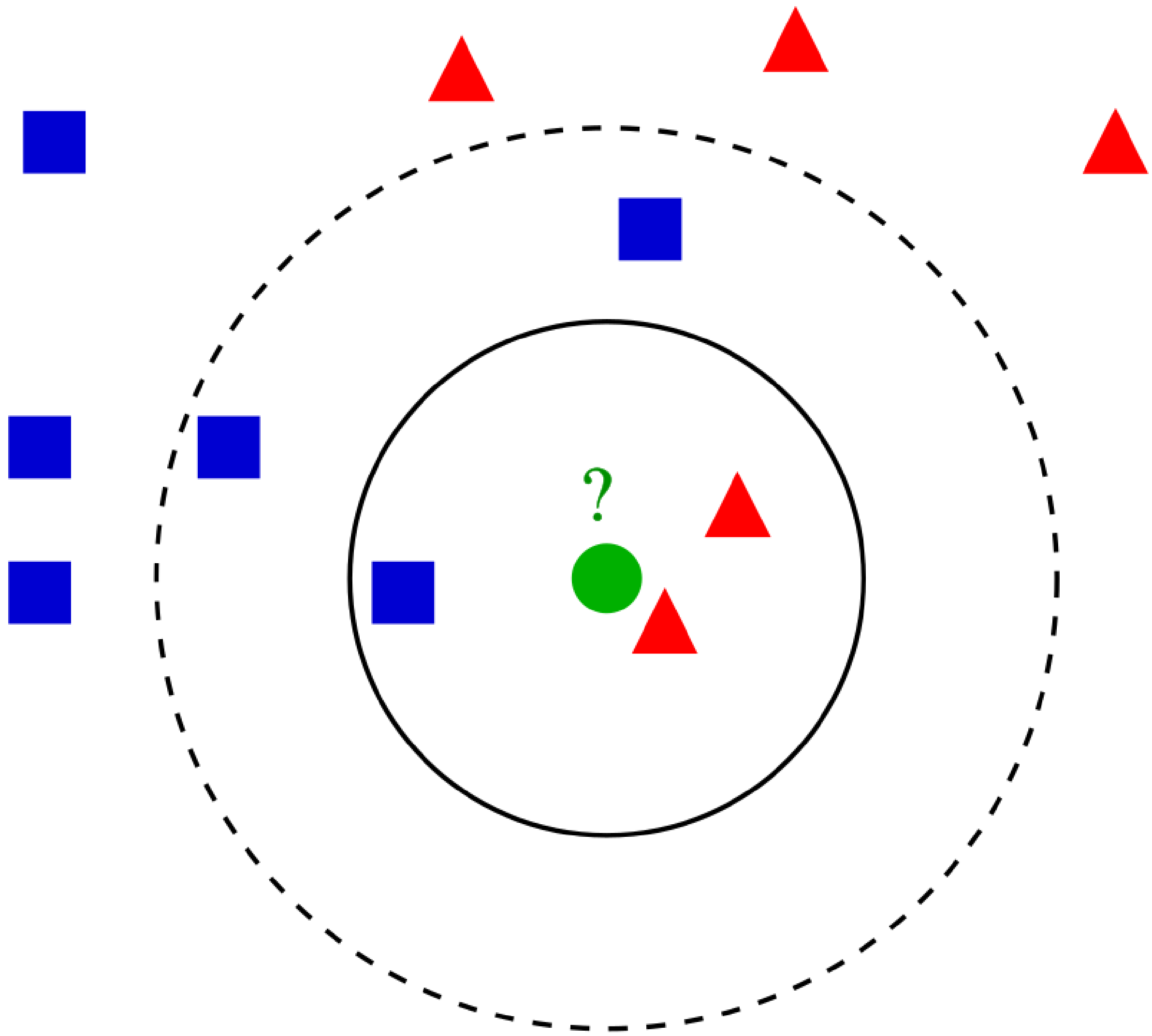


Figure 4: kNN Spatial Geometry

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

This is due to its allowance for case-based reasoning (e.g., explaining that an employee is at risk because they share similarities with 5 specific former employees who terminated). The number of nearest neighbors used for prediction (k) is obtained empirically and varies by dataset structure.

6. **Logit Link Function (Generalized Additive Models - Model F):** Generalized Additive Models (GAMs) are introduced to the predictive panel to bridge the gap between rigid linear models and their blackbox counterparts in handling nonlinearity robustly. This robustness is materialized in GAMs ability to replace standard linear coefficients with non-linear "smoothing splines," allowing the model to capture the natural ebbs and flows in HR datasets which is secured through a flexible, piecewise polynomial function employed to model nonlinear relationships between variables by fitting smooth, joined segments of data.

While GAMs enjoy a multiplicity of configurable options in the form of link functions (called so because their responsibility in relating the expected value of the response variable to the additive predictor thereby transforming nonlinear relationships into a linear scale), the implementation of the Logit link function is uniquely suited to the binary outcome variable being modeled in this study (attrition as a 1 or 0 flag). Similar to logistic regression, Logit link GAMs models output probabilities which lend themselves well to digestibility of model output in addition to permitting edge cases for SHAPs analysis (the employee records with highest and lowest probabilities of attrition). GAMs models however, contrast themselves from Logistic regression models in their computational approach in generating probabilities as these models calculate probabilities using the sum of these smooth functions (or splines) rather than a rigid linear combination of variables.

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + s_1(x_1) + s_2(x_2) + \cdots + s_n(x_n)$$

GAMs models further enrich this study with their delivery of Partial Dependence Plots (PDPs) endemic to their conceptual architecture which allow researchers to visually isolate the exact inflection points where flight risk spikes (e.g., specific tenure milestones) , offering interpretability that standard models are otherwise liable to miss.

Given that the catalogue of features between the three datasets used in this study is expansive global Variable Importance Measures (VIMs) will first at first isolate the most highly influential features for subsequent exploration via PDPs to balance the computational efficiency of the overall pipeline.

2.3 Experimental Design and Analytical Procedures

In congruence with standard best practices in machine learning methodology this study will employ an 85/15 train-test split employed across all datasets to alleviate the danger of overfitting the data at hand. Additionally, in paying fidelity to the protocols of Mantovani et al. [14], tools like `RandomizedSearchCV` from the scikitlearn library are used to iterate through hyperparameter grids to find the mathematically optimal configuration before final predictions.

Furthermore, to properly remediate the susceptibility of ML models to conform to class imbalanced datasets (which are chronically observed in enterprise HR settings while modeling attrition), class weights are computed to synthetically penalize the algorithms heavier for missing a terminated employee thereby liberating these models from lazily predicting universal retention of employees.

The evaluation will be grounded on an array of diagnostic techniques discussed below that will be applied to all the above models where applicable:

- **The Case for Recall:** The use of Accuracy as a primary metric in attrition studies often mars results by skewing the predictive model towards the attributes of the dominant class of the classification model. A turnover classification model simply predicting that "every employee stays" could achieve an accuracy rate of ~84% (for example) due to the high class imbalance pervasive in HR datasets. This slight of hand in classification model interpretation is termed as a "no-information rate".

$$Recall = \frac{TP}{TP + FN}$$

  Therefore, Recall (the percentage of actual leavers a model correctly identifies) justifies itself as a potent metric under the auspices of Workforce Management as it compels the model to prevaricate missed retention opportunities.

- **F1-Score:** As Recall and Precision are often inversely related, the inclusion of the F1-Score offers us a balanced comprehension for the model's overall capabilities and trade-offs with respect to performance.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

- **Elbow Diagram:** A diagnostic solution particular to kNN algorithms, the Elbow Diagram plots Model Error (Y-axis) against the number of neighbors k (X-axis). The "Elbow" occurs where the marginal gain in predictive power diminishes. The optimal k is selected at this inflection point (e.g. if the error rate stabilizes at $k = 4$, then 4 neighbors are selected to balance complexity and bias). Unlike tabular metrics that provide direct, localized maxima or minima, the elbow method utilizes a graphical heuristic to identify the point of stabilization in the error rate.

- **Variable Importance Measures (VIMs):** In computing this metric, relative scores are assigned to all model features, representing their proportional importance with respect to the most influential variable. These scores assist in isolating Global turnover factors that can be primed for HR intervention (e.g., identifying if `OverTime` is the dominant driver across the entire organization).

- **Shapley Values (SHAP):** This is a Localized metric used to diagnose why a specific individual was flagged.

$$\phi_i(val) = \sum_{S \subseteq \{1,\dots,p\} \setminus \{i\}} \frac{|S|!(p-|S|-1)!}{p!}(val(S \cup \{i\}) - val(S))$$

  SHAP values are delivered as case-wise statistics that delineate the contribution of a particular predictor (eg, `DistanceFromHome`) taken against the baseline. Due to the granular nature of these metrics, we will confine their use to specific edge cases, just as Wu & Fukui [13] did.

To synthesize the theoretical frameworks and evaluative metrics discussed throughout this section into a cohesive operational workflow, the execution of the MARS-Machine Learning pipeline is encapsulated in the procedural logic below. This algorithmic summary functions as the definitive computational blueprint for the experiments conducted in the subsequent phases of this study:

```
# ALGORITHM 1: MARS-Machine Learning Predictive Pipeline

INPUT: Datasets D (IBM, Babushkin turnover dataset, Data Scientist Job Change dataset)
OUTPUT: Evaluated Models (A-F), Classification Summary Reports,
Variable Importances, PDPs, SHAP Values

1. PREPROCESSING:
    Apply 85/15 Train-Test Split -> (X_train, X_test, y_train, y_test)
    Compute Class_Weights to penalize False Negatives (Attrition = 1)

2. PIPELINE EXECUTION:
    FOR EACH Model_Type IN (Model A, B, C, D, E, F):

        # Hyperparameter Optimization (Mantovani Protocol)
        Define Grid_Space (G)
        Best_Model = RandomizedSearchCV(Model_Type, G, X_train, Class_Weights)

        # Prediction & Evaluation
        y_pred = Best_Model.predict(X_test)
        Generate Confusion_Matrix(y_test, y_pred)
        Calculate Recall, F1_Score

      # Global Interpretation
        IF Model == Decision Tree (Model D):
            Extract Gini_Importance -> Plot Top Predictors
```

```
    ELSE IF Model == kNN (Model E):
        Extract SelectKBest_F_Scores -> Plot Top Predictors
    ELSE IF Model == GAM (Model F):
        Extract VIMs -> Construct Partial Dependence Plots (PDPs)
    ELSE (Linear/Logistic Models):
        Extract Beta_Coefficients -> Plot Top Predictors

    # Local Interpretation (Edge Cases)
    IF Model outputs Probabilities (Logistic, GAM) OR Geometric Margins (SVM):
        Identify Edge_Case_Indices (Highest/Lowest risk bounds)
        Calculate SHAP_Values(Best_Model, Edge_Case) -> Generate Waterfall Plot

END FOR
```

2.4 Software and Tools

Python 3 will function as the computational engine of this study with the preponderance of data ingestion, manipulation, and array handling (like dealing with the Babushkin dataset's sparsity) executed in the `pandas` and `numpy` libraries.

The computation of the Classification models fashioned in this pipeline will be borne by Python's `scikit-learn` ecosystem which will be integral to the construction of Models A through E. The grid search and scaling techniques discussed elsewhere in this section are also the progeny of `scikit-learn` libraries.

Renderings of the visual array of diagnostic plots like Shapely plots (derived via the `shap` library), VIMs feature plots and elbow diagrams operationalized by this study will emanate from `matplotlib` and `seaborn` libraries, with Model F and its corresponding PDP plots being sourced from the `pygam` library.

Finally, all computations herein are executed locally utilizing standard CPU resources via a Jupyter/Quarto environment.

2.5 Ethical Considerations and Bias Mitigation

Enterprise HR data is among the most sensitive flavors of data produced by most corporations due to an abundance of demographic information like race, sex and age as well as compensatory information like wage and bonus information. Hence, the three principal datasets used in this study have been scrubbed of Personally Identifiable Information (PII) prior to ingestion to sew up baseline confidentiality.

Additionally, the experimental design implemented in this study rejects such variables implicitly due to the general inapplicability of demographic variables to MARS factors, and thereby this demographic-agnostic approach serves as a structural safeguard for privacy and prevents algorithms from anchoring on protected classes.

The deliberate choice of interpretable models assembled in this study (Models A through F) and record-level post-hoc diagnostics like SHAP illuminate HR stakeholders as to why a flight risk flag was generated and consequently ensures retention interventions are equitable and guarded from hidden algorithmic bias.

3 Capstone Results & Discussion

3.1 Results: IBM Dataset

Here the IBM dataset is pruned to conform to the MARS paradigm, given that the data was heavily skewed towards retiention (i.e the target variable had many more instances of retention than attrition), a Python dictonary of class weights is created to force the model's hand in prioritizing attrition. While non-informative columns are dropped as they were with the IBM dataset in this study, the native `profession` column from the Babushkin dataset (which is a multi factor categorical column indicating the specialization of an employee) will be tranformed into the binary `hr_profession` variable as exempting this variable from transformation will have compelled the models in this pipeline to split on highly sparse, fragmented categories for the remaining 50% of the company not engaged in HR roles thereby diluting the predictive power of the feature considerably. Hence this transformation forces the analytical pipeline to confront a highly specific binary question: Does being an HR insider structurally alter an employee's retainability by the organizations that employ them?. The forthcoming sections will serve to answer this query.

```
IBM Dataset Pipeline | Shape Filtered: (1470, 35) -> (1470, 16)
Computed Imbalance Weights: {0: 0.6, 1: 3.1}

Final MARS-Tagged Features:
 - JobSatisfaction (M)
 - TrainingTimesLastYear (M)
 - StockOptionLevel (M)
 - PerformanceRating (A)
 - Education (A)
 - YearsAtCompany (A)
 - JobLevel (R)
 - YearsWithCurrManager (R)
 - YearsInCurrentRole (R)
 - DistanceFromHome (S)
 - EnvironmentSatisfaction (S)
 - EducationField_Human Resources (A)
 - EducationField_Life Sciences (A)
 - EducationField_Marketing (A)
 - EducationField_Medical (A)
 - EducationField_Other (A)
 - EducationField_Technical Degree (A)
 - Department_Human Resources (R)
 - Department_Research & Development (R)
 - Department_Sales (R)
```

```
- OverTime_0 (S)
- OverTime_1 (S)
- BusinessTravel_0 (S)
- BusinessTravel_1 (S)
- BusinessTravel_2 (S)
```

3.1.1 Confusion Matrix Display & Analysis (IBM Dataset)

Figure 5 below represents a **confusion matrix** which is a visualzation that employs a square N x N marix summarizing a **Classification model**'s performance by mapping true class labels against their predicted counterparts obtained from the **classifiers**. It also allows for calculating key **performance metrics** that were discussed in the Methods section of this study like **precision**, **recall**, and F_1 **scores** to identify class-specific performance issues.

As such, Figure 5 will serve as an illustrative example of how these key **metrics** are obtained with this tecnique while future computations of these **scores** will be tabulated in summary tables to enhance interpretability and maintain control over visual overload.In terms of the analytical utility of Figure 5, it can be observed that the results gleaned in this visual serve as a teaching example of the "**Accuracy** Trap" pervasive in **Machine learning** and *attrition* studies, The **Std. Logistic Regression model** here misclassified 28 out of 36 *attriting* employees as being *retainable* while providing a reasonbaly **accuracy rate** (~85%) that could confound the naive interpreter into overstating the utility of this **model** to drive *retention* in enterprise HR settings.

This failure at once justifies the use of **regularization techniques** in **linear models** and the inclusion of **distance based** and **non-linear models** in our overall methodology to offset these observed eccentricities.

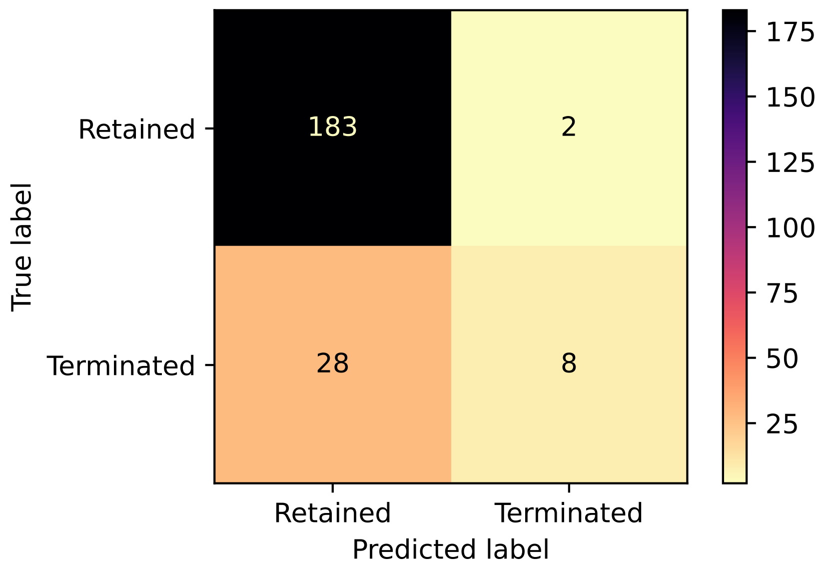


Figure 5: Confusion Matrix for Std Logistic Regression Model (IBM)

```
The classification report for the A: Standard Logistic Regression model is:
              precision    recall  f1-score   support

           0       0.87      0.99      0.92       185
           1       0.80      0.22      0.35        36

    accuracy                           0.86       221
   macro avg       0.83      0.61      0.64       221
weighted avg       0.86      0.86      0.83       221
```

3.1.2 kNN Elbow Diagram & Feature Selection (IBM Dataset)

kNN Classifiers are notoriously prone to having their geometiric distance calculations diluted by an overabundance of **model** features, therefore sklearn's `SelectKBest` will be used to shrink our featureset down to the 10 most influential variables, this is accomplished by by performing a one-way **Analysis of Variance (ANOVA) F-test** (by using the `f_classif` setting of this function) between each individual numerical feature and the categorical **target variable** to determine their linear dependency and then narrowing down the feature set by selecting Features with the highest resulting **F-statistic scores**.

```
kNN will now use only these 10 features:
['JobSatisfaction (M)' 'StockOptionLevel (M)' 'YearsAtCompany (A)'
 'JobLevel (R)' 'YearsWithCurrManager (R)' 'YearsInCurrentRole (R)'
 'EnvironmentSatisfaction (S)' 'OverTime_0 (S)' 'OverTime_1 (S)'
 'BusinessTravel_2 (S)']
```

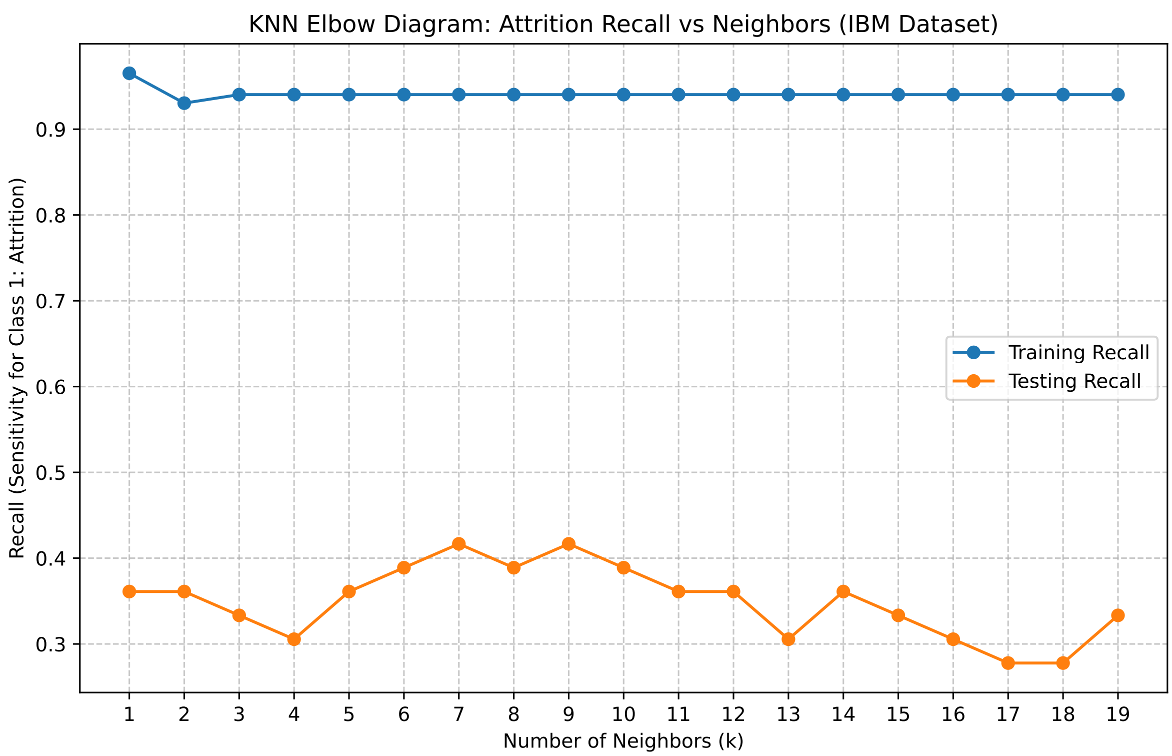


Figure 6: KNN Elbow Diagram: Attrition Recall vs. Neighbors (IBM Dataset)

```
Peak Performance: k=7 with Test Recall of 0.4167
```

Observing Figure 6 above, we discover that the **kNN algorithim** (**Model E**) identified **k = 7** as the optimal number of neighbhors, which in the realm of enterprise HR translates to an employee's *flight risk* being best obtained from a "pod" or circle of their 7 closest peers.

3.1.3 Global Feature Importances (VIMS Facet Grid - IBM Dataset)

While the **kNN model** above computes its insights via the construction of peer groups, it is imperitive for us to understand the Global implications of the features in overall dataset in affecting *turnover outcomes*, this is accomplished via the following facet grid of feature importances Figure 7

Readily apparent from Figure 7 above, are the elevation of `JobLevel (R)` and `YearsWithCurrManager (R)` features as universal predictors of *retention* (indicated by their representation as blue bars) via our panel of **predictive models**. Conversely OverTime_1 (S) (or being called into *Overtime shifts/labor*) is universally flagged as a top *Attrition Driver* (Red bars) or a top magnitude splitter (Grey bars).

This prioritization underscores the very limited utility of *financial or compensatory recourse* to alleviate *attrition* emanating from poorly defined roles (low values of `JobLevel (R)`) or poor or experimental managerial allotment and configuration (demonstrated by low values of `YearsWithCurrManager (R)`)

Also visible here is the impact of **regularization techniques** that reveal themsevles in the disparity betweeen the feature importances produced by **Models A & B**. In **Model A** `YearsAtCompany (A)` is a moderate red bar and the #9 overall factor in predicting *attrition*, however in **Model B** this feature ascends in importantance to the #1 predictor of *attrition* ostensibly due to the preponderance of multicollinear or correlated feautes that are endemic to HR datasets (e.g *tenure*, *time in role* etc). In penalizing this noise with an **L2 penalty**, **Model B** uncovers a harsh HR reality: Working at an organization for extended periods of time without progressing upwards in terms of *Job Level* or *seniority* or being transferred to a different manager leads to *stagnation* and *flight-risk*. In short: Decoupling *tenure* from *job mobility* can have deleterious consequences for *retention*.

The **decision tree** (**Model D**) offers a unique understanding of *turnover trends* in IBM in its promotion of `OverTime_0` and `JobLevel` as predictive variables, this promotion is computational rather than accidental.*Decision trees **eschew constructing gradual slopes or** probabilities **in their endeavor to mine insights but instead use** metrics of impurity (gini **or** log-loss**) to find the cleanest cuts in bifuricating the workforce into *stayers* and *leavers*. This is most operational when HR managers require hard-and-fast operational rules to flag *flight risks* (rather than gradual** probabilities**) and in this vein** Model D** avers that overtime status (or the lack therof) provides the cleanest immediate filter.

Non-linear models in the Figure 7 panel, such as the **Logistic GAM (Model F)**, underscore the primacy of targeted retention efforts like stock grants (denoted by `StockOptionLevel (M)`)

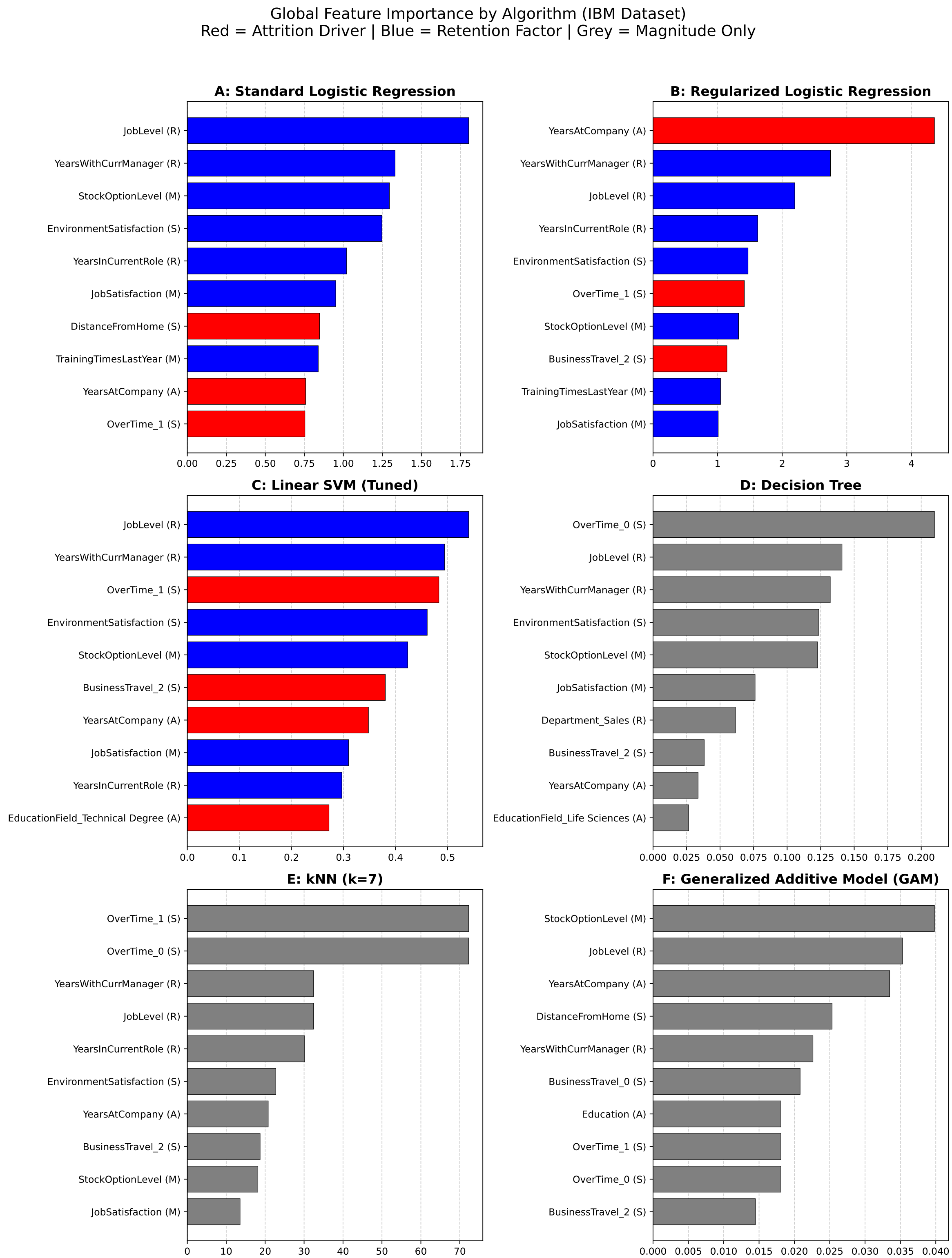


Figure 7: Global Feature Importance by Algorithm (IBM Dataset)

in the IBM dataset, despite only being marginally influential in the linear models of the panel (**Models A through C**). This is likely due to the nature of linear models in forcing straight lines through the dataset; doing so in the context of **`StockOptionLevel (M)`** implies that every extra stock option confers upon the enterprise the *exact same amount of loyalty*. GAMs dispel this illusion and allow for a nuanced understanding with their usage of splines. The resulting implication here is that extrinsic motivators like compensatory rewards are not flat and can experience *diminishing returns* beyond a certain threshold, which will be explored further in the next section.

3.1.4 Logistic GAM Partial Dependence Plots (IBM Dataset)

he **Partial Dependence Plots (PDPs)** rendered below isolate a single variable to show its exact, marginal effect on an employee's **probability** of quitting, holding all other variables constant. Within each **PDP** diagram, the X-axis represents the employee's status for that specific feature, scaled from 0.0 (lowest possible value in the dataset) to 1.0 (highest possible value). The Y axis represents the **log-odds** of our binary **target variable** (*Attrition* for the datasets employed in this study), most crucial here is the horizontal dashed line at 0.0 which can be understood in lay terms as the "*turnover line*", therefore any point on the curve above the zero line means that specific value drives the employee to depart the enterprise (higher *attrition risk*) wheras Any point below the zero-line means that value drives them to stay by acting as a *retention anchor*.

Accordingly, the **PDP** plots above in Figure 8 offer us validation in the previously concluded limited utility of *compensatory rewards*, the `StockOptionLevel (M)` plot demonstrates here that conferring a mid-tier *stock grant* on an employee buys the issuing enterprise massive *retention power* but issuing the absolute maximum *stock grant* value squanders entrpise capital in its failure to secure additional *loyalty* in proportion to the elevated stock issuance.

The `JobLevel` **PDP** plot adds to our nonlinear grasp of IBM data by highlighting the prevalance of a "*Middle-management squeeze*" at IBM, *Attrition risk* starts out high for entry level employees (`JobLevel` = 0.0) drops below the zero line before again resurfacing at a middle range (`JobLevel` between 0.4 and 0.5) before its terminal plunge into *retentive safety* for Senior executives (`JobLevel` >= 0.8).This insight escaped the scrutiny of the **linear models** because they'd averged out the very same **splines** that produce these insights.**Logistic GAM** reveals here that middle managers at IBM are a considerable *flight risk* ostensibly due to them coming out on the losing end of the *stress to compensation ratio* (i.e they have comparable *workload* to senior leadership sans the lucrative *stock rewards* or the enhanced *autonomy* prevelant at the upper eschelons of an enterprise)

The `YearsAtCompany` **PDP** plot offers its own cautionary tale in the form of demonstrable *tenure fatigue*. It can be observed in Figure 8 that the curve stays below the zero line for people who are in the early or intermediate stages of their *career trajectory* (`YearsAtCompany` between 0.0 and 0.7), but once this threshold is crossed, the **log-odds** favoring *attrition* rocket upward

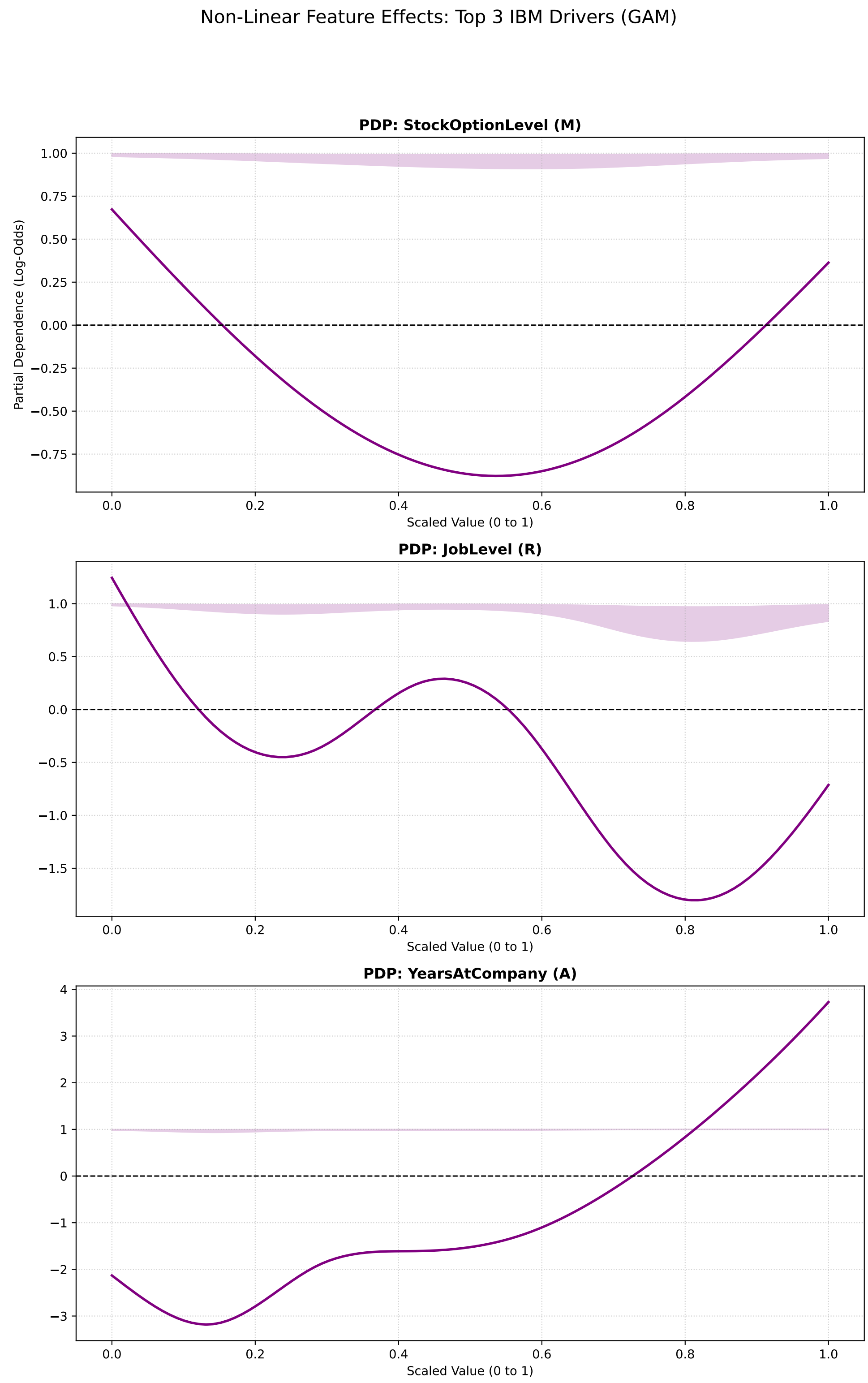


Figure 8: Non-Linear Feature Effects for Top Continuous Drivers (IBM GAM)

exponentially.Translated to Enterprise HR strategy, this insight informs us that *Institutional loyalty* has an expiration date in that employees may very well be *retained* comfortably by an enterprise for several years but if they arrive at the late stages of their *tenure* without haviung been promoted or structurally re-engaged, "*tenure fatigue*" is likely to take hold and exponetiate their *attrition risk*.

While the **PDP plots** from the **GAM model** offer tremendous theoretical clarity in the full spectrum observation of *workforce behavior*, their practical limitations in implementation (e.g., an HR manager cannot meaningfully deploy a "curve" into an automated alert system) necessitate the hard binary logic of **Decision Trees** to locate specific "*Kill Zones*" of *turnover* in an effort to bridge the gap between *behavioral theory* and operational HR intervention.

3.1.5 Decision Tree Attrition "Kill-Zones" (IBM Dataset)

```
Kill Zone 1: 92.8% Attrition Concentration
Impact: 0.9/1.0 weighted score (Applies to 35 actual employees | 2.80% of workforce)
Logic Path: OverTime_0 (S) <= 0.50 AND JobLevel (R) <= 1.50 AND YearsInCurrentRole
(R) <= 0.50

Kill Zone 2: 90.6% Attrition Concentration
Impact: 0.9/1.0 weighted score (Applies to 20 actual employees | 1.60% of workforce)
Logic Path: OverTime_0 (S) <= 0.50 AND JobLevel (R) <= 1.50 AND YearsInCurrentRole
(R) > 0.50 AND DistanceFromHome (S) > 12.50

Kill Zone 3: 89.4% Attrition Concentration
Impact: 0.9/1.0 weighted score (Applies to 21 actual employees | 1.68% of workforce)
Logic Path: OverTime_0 (S) <= 0.50 AND JobLevel (R) > 1.50 AND Department_Sales (R) >
0.50 AND StockOptionLevel (M) <= 0.50 AND DistanceFromHome (S) > 5.50
```

he cohort specified by Kill Zone 1 is comprised of Junior employees (`JobLevel` <= 1.50) brand new to their role (`YearsInCurrentRole` <= 0.50) who are recruited into working *overtime* (`OverTime_0` <= 0.50,`OverTime_0` must be equal to 1 to indicate the absence of *overtime*). In an HR context this stipulates that if IBM hires new employees into brand new roles and forces them to work beyond business hours this will come at the cost of 93% percent of this cohort *attriting*, these employees would *burn out* before their *onboarding* is even completed in earnest.

Kill Zone 2 represents a class of more tenured Junior employees (`JobLevel (R)` <= 1.50 AND `YearsInCurrentRole (R)` > 0.50) than in Kill zone 1, like the employees in Kill Zone 1, these employees also work *overtime* (`OverTime_0` <= 0.50) however they are uniquely beleaguered by long *commutes* to the office (`DistanceFromHome` > 12.50). The curative HR remedy to this issue would be thhe facilitation of additional *remote work* days or transportation stipends to alleviate the physical *exhaustion* faced by Junior employees who struggle with both extended working ours and arduous *commute* journeys.

Kill Zone 3 offers a unique look at the vulnerabilities of organizational *veterancy* particularly in *Sales roles* (`JobLevel (R)` > 1.50 AND `Department_Sales (R)` > 0.50), this cohort also suffers from compromised *work life balance* (OverTime_0 (S) <= 0.50) as well as the limitations of IBM's reward structure (`StockOptionLevel (M)` <= 0.50).This logical path points to classic *sales burnout* obseved in enterprises all over the world. Experienced sales representatives are being compelled to grind beyond business hours without long-term *extrinsic rewards* to lock in their *committment*. Absent this *compensatory equity*, veteran sales staff will simply take their client books to a competitor.

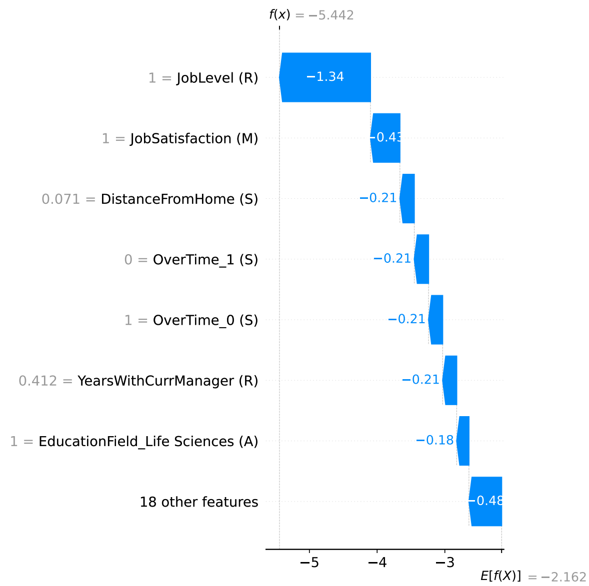
While **Decision Trees**, as illustrated by the 3 Kill-Zones, can produce watershed understandings of high-risk groups within an organization, HR departments ultimately concern themselves with the well-being of individuals. Managers need to know with specificity which of the employees they manage is a *flight risk*. To address this need, this study employs **SHAP Extreme Case Analysis** in the next section as the final proof of concept by demonstrating that the pipeline can surgically diagnose the distinct mathematical pressures acting on a single employee.

3.1.6 SHAP Extreme Case Analysis: The Safe Bets & *Flight Risks* at IBM

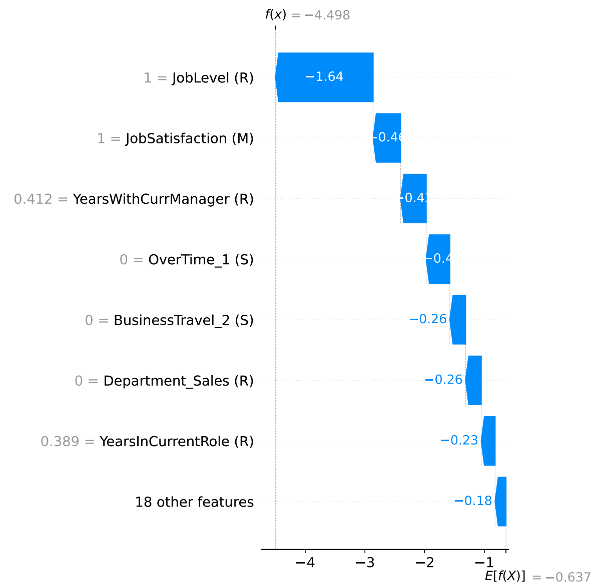
Below in Figure 9, we notice that the massive blue bars pushing the employee's risk profile into the negative (safe/*retention*) territory are `JobLevel` and `YearsWithCurrManager`.The global insights observed in Section 3.1.3 are validated here at the individual level in our discovery that the most *retainable* people at IBM aren't necessarily the most generously *compensated* but instead employees grounded in a deep structural *seniority* by virtue of their roles in the organization and also enjoy long standing positive *relationships with their managers*.

The riskiest employees as seen in Figure 10 are those plagued by forced *overtime* (`OverTime_1`) and critical of their environmental situation at the organization (`DistanceFromHome`).Herein lies the surgical power of **SHAP analysis** as it can be inferred from this diagnostic panel (Figure 10) that a successful *retention strategy* for these high-risk employees must manifest as schedule and *workload relief* as opposed to incremental *compensatory bonuses* like salary raises.

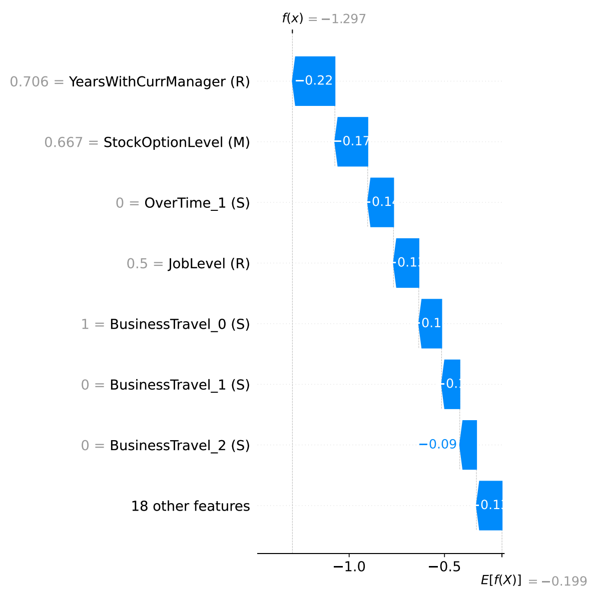
Our pipeline thus far as implemented in Section 3.1 successfully proves that highly accurate and indivudalistic diagnostics can be just as efficacious as those produced by "**black-box**" **models**. Hence, with the theoretical and diagnostic capabilities of the **models** proven, the final step in the IBM evaluation is an empirical ranking and tabulation to determine which **algorithm** actually caught the most *flight risks*.



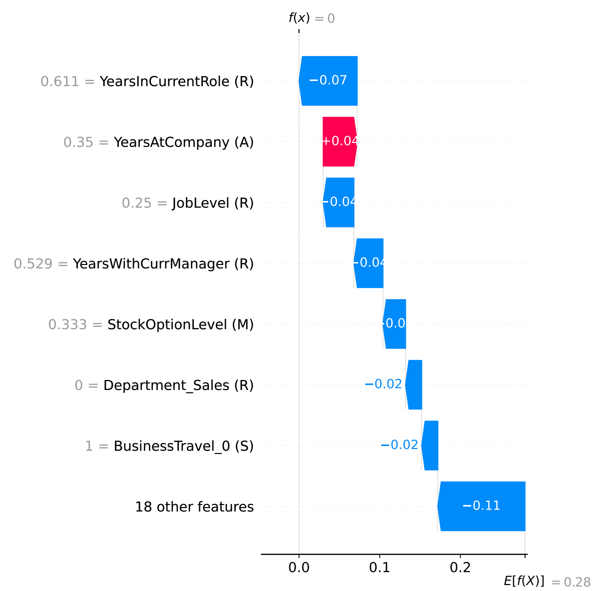
(a) A: Std Logistic Regression - Safest



(b) B: Reg. Logistic Regression - Safest

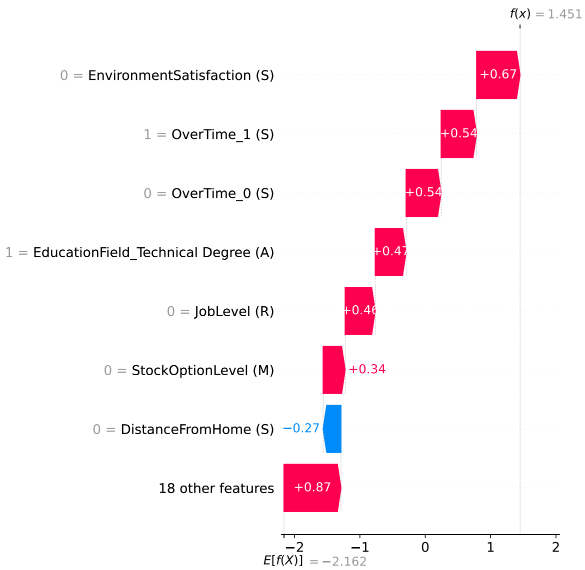


(c) C: Linear SVM (Tuned) - Safest

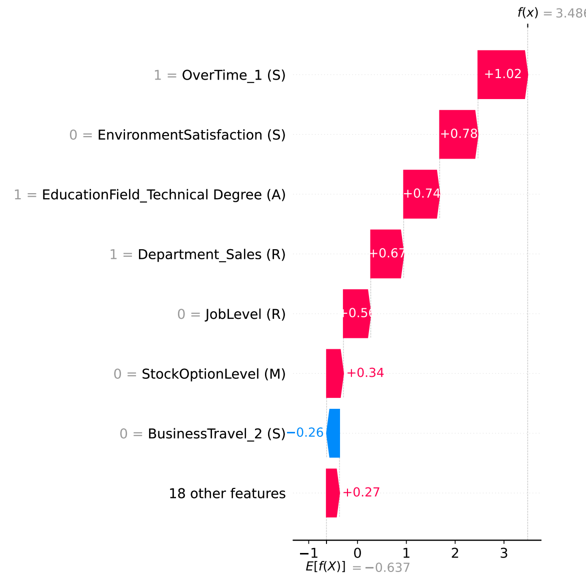


(d) F: GAM - Safest

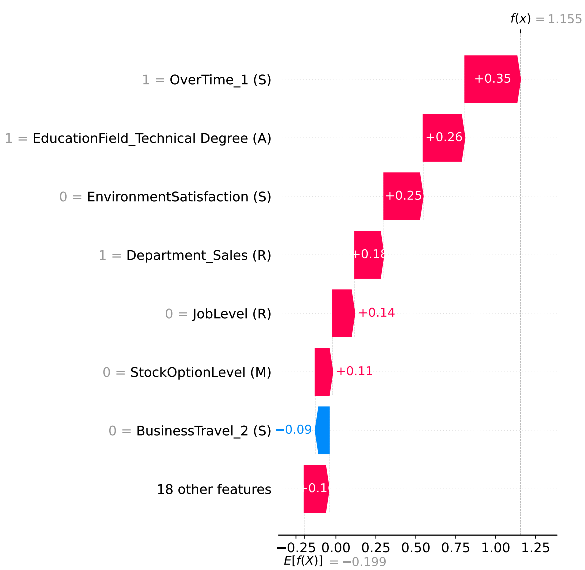
Figure 9: SHAP Local Interpretability: Profiles with Lowest Attrition Probability (IBM)



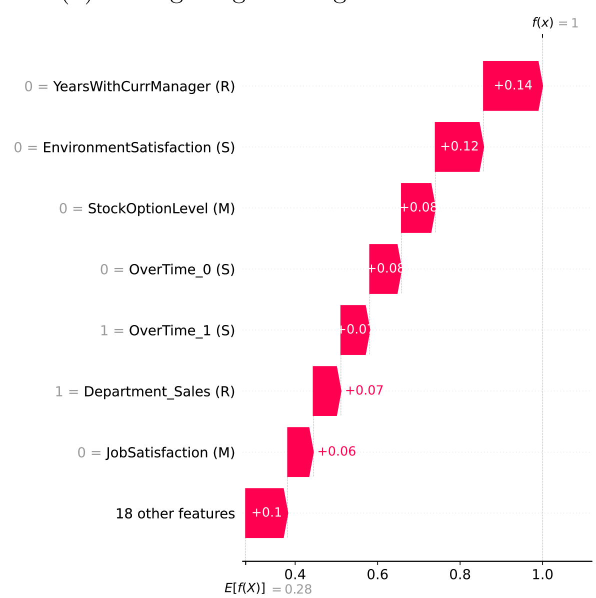
(a) A: Std Logistic Regression - Riskiest



(b) B: Reg. Logistic Regression - Riskiest



(c) C: Linear SVM (Tuned) - Riskiest



(d) F: GAM - Riskiest

Figure 10: SHAP Local Interpretability: Profiles with Highest Attrition Probability (IBM)

3.1.7 Model Performance Matrix (IBM Dataset)

Table 1: Consolidated Algorithm Performance and Hyperparameters (IBM Dataset)

Rank	model name	tuned hyperparameters	Retention Recall (Class 0)	Attrition Recall (Class 1)	F1-Score (Weighted)
1	D: Decision Tree	{'min_samples_split': 2, 'min_samples_leaf': 20, 'max_depth': 5, 'criterion': 'log_loss'}	0.681	0.75	0.731
2	F: Generalized Additive Model (GAM)	{'lam': 0.1}	0.795	0.722	0.804
3	C: Linear SVM (Tuned)	{'C': 0.0379269019073225}	0.741	0.694	0.764
3	B: Regularized Logistic Regression	{'solver': 'saga', 'penalty': 'l2', 'C': 555.556}	0.735	0.694	0.76
5	E: k-Nearest Neighbors	n_neighbors=7	0.957	0.417	0.857
6	A: Standard Logistic Regression	Baseline (Untuned)	0.989	0.222	0.83

We see in Table 1, the **Decision Tree** performs best with the primary **evaluation metric** of this study by securing an **Attrition Recall rate** of 0.75 implying that the **algorithm** successfully identifies 3 in 4 employees that actually terminated, this is in stark contrast to the untuned **baseline model** (**Model A: Std Logistic Regression**) which with an **Attrition recall** of 0.22 could only muster the successful identification of 1 in 5 *leavers* which cements **Model D** as a substantial computational upgrade to this baseline.

The best holistic compromise however is **Model F** (**Logistic GAM**) which earned a near comparable primary **metric score** of 0.722 to **Model D**'s 0.75, but maintained a much higher F_1 **score** than its competitor in **Model D** ($F_1 = 0.724$), this is due to its superior performance in the domain of **retention recall**. The enterprise HR implication of this tabulation therfore avers that while **Model D** flags the most *flight-risks*, **Model F** offers comparable performance for fewer tradeoffs as it avoids burying HR in false-positive *retention* meetings.

Hence Table 1 validates the introductory hypothesis of this study that the evolution of data pipelines from rigid, **lineal algorithms** (**Models A through C**) into inherently transparent and interpretable **non linear models** like **GAM** and **Decision trees** not only provided superior business intelligence but also yielded the highest absolute **predictive performance**.

3.2 Results: Babushkin Dataset

Because the Babushkin dataset was synthtecially compiled as a **balanced dataset** (there is parity between records indicating retention and termination), Thr computed class weights are {Retention: 1.01, Attrition: 0.99} owing to the abscence of a massive class imbalance in the dataset to skew the baseline.

The Babushkin dataset sees an overall increase in features inspite of dropping non-informative and demographic columns due to the *one-hot encoding technique* producing multiple binary variations of the categorical columns from the native unmodified dataset.

```
Babushkin Dataset Pipeline | Shape Filtered: (1129, 16) -> (1129, 22)
Computed Imbalance Weights: {0: 1.01, 1: 0.99}

Final MARS-Tagged Features:
 - stag (A)
 - greywage (S)
 - extraversion (M)
 - anxiety (M)
 - selfcontrol (M)
 - novator (M)
 - independ (M)
 - hr_profession (R)
 - coach_my head (R)
 - coach_no (R)
 - coach_yes (R)
 - way_bus (S)
 - way_car (S)
 - way_foot (S)
 - traffic_Agency_Ads (S)
 - traffic_Job_Boards (S)
 - traffic_Referral_Network (S)
 - industry_Heavy_Industry (R)
 - industry_Public_Sector (R)
 - industry_Retail_Services (R)
 - industry_Tech_Professional (R)
```

3.2.1 kNN Elbow Diagram & Feature Selection (Babushkin Dataset)

We observe here in the tuning of the **kNN model** that the **ANOVA F-test** employed by the Python **`SelectKBest`** function hasv iolently pruned the *Five-Factor Model (FFM) personality traits.* The features **`anxiety (M)`** (corresponds to *Neuroticism*) and **`independ (M)`** (corresponds to *Openess to Experience*) were the only two factors of the *Five-Factor Model* that met the threshold of **linear dependency** with the target variable, this selection makes salient the limited utility of *extensive personality screening* as a procedural step in *enterprise hiring practices*.

```
kNN will now use only these 10 features:
['stag (A)' 'anxiety (M)' 'independ (M)' 'hr_profession (R)'
 'coach_yes (R)' 'way_bus (S)' 'way_foot (S)' 'traffic_Job_Boards (S)'
 'traffic_Referral_Network (S)' 'industry_Public_Sector (R)']
```

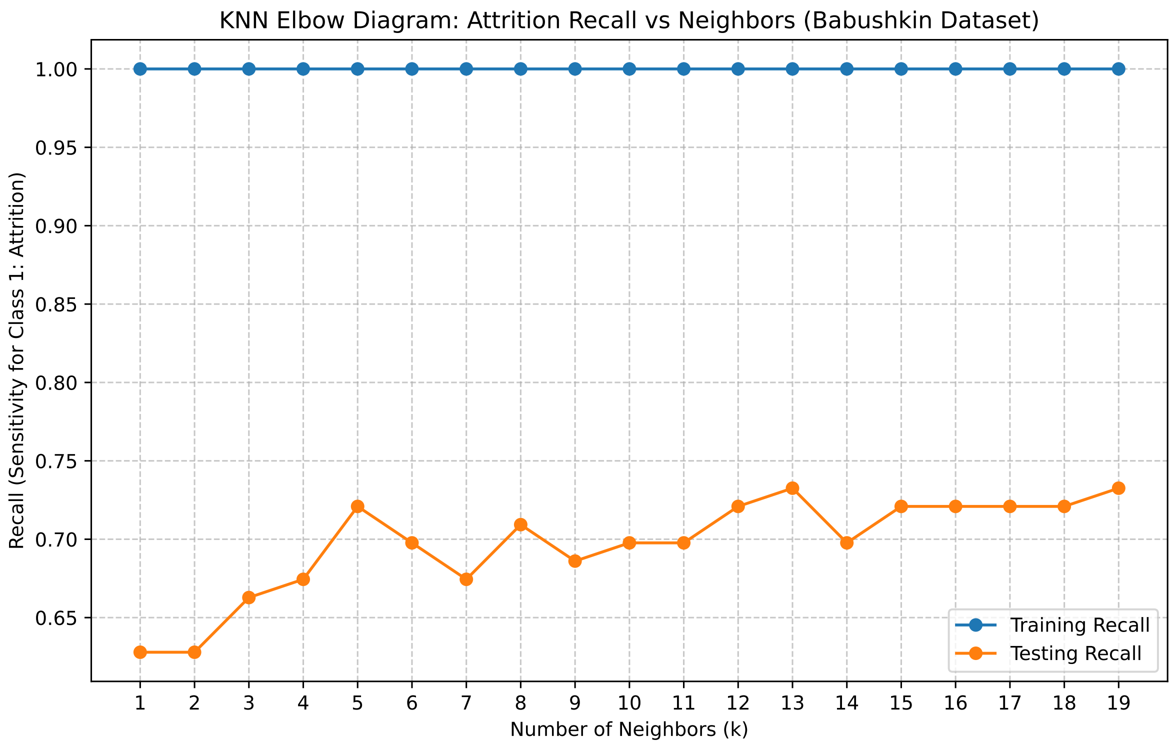


Figure 11: KNN Elbow Diagram: Attrition Recall vs. Neighbors (Babushkin Dataset)

```
Manually Selected for Interpretability: k=5 with Test Recall of 0.7209
```

Though the zenith of **recall** in Figure 11 is met only at $k = 13$ or higher, a manual override of $k = 5$ is made to ground the implementation of this algorithim in *business practicality*: While HR managers can meanigfully oversee and intervene in a *localized pod or team* of the employee's 5 closest peers, allowing the algorithm to optimize blindly and demand a *peer group* of 13 or more people would compromise its servicability as *peer groups* this large exceed *organizational spans of control* that most humans are capable of overseeing.

3.2.2 Global Feature Importances (VIMS Facet Grid - Babushkin Dataset)

Particularly arresting about Figure 12 above is the contrast between both **Logistic Regression** models (**Models A & B**), while **Model A** insists that **`anxiety (M)`** and **`stag (A)`**(tenure) are dominant in ascertaining *flight risk*, **Model B** favors Role based and situational factors like **`hr_profession (R)`** and **`way_foot (S)`**. This occurs due to the **L2 penalization** implemented by **Model B** which eradicates **`anxiety (M)`** and deprioritizes **`stag (A)`**. This informs us that the baseline model was taken in by correlated noise but by the day to day realities of their incumbent role (**`hr_profession (R)`**) and the logistical ease of their commute (walking to work acting as a massive *retention anchor* via the **`way_foot (S)`**)

A noticable drop off in feature importance also presents itself in Figure 12 with **Model D (Decision Tree)**, while the **linear models** distribute importance across a wide curve, the **decision tree** restricts its prioritization to merely four variables with **`stag (A)`** topping the list followed distantly by **`traffic_Job_Boards (S)`**, **`selfcontrol (M)`**, and **`independ (M)`**. This in part due to the computational nature of **decision trees** in being liberated from charting gradual psychological probabilities or drawing straight lines. Metrics like **Gini impurity** or **log-loss** are instead operationalized to aggressively hunt for the single cleanest mathematical cut that divides stayers from leavers. This suggests that if HR managers are to fashion a hard *binary alert system* to flag turnover, they would be well advised to ignore psychological surveys entirely. **Model D** champions **`stag (A)`** as the most mathematically pure threshold to flag a *flight risk*, making almost all other features computationally irrelevant for the initial operational filter.

Gleaning at Figure 12 above in a more wholistic sense, an observable consensus between the **linear models** emerges that **`anxiety (M)`** is the most crucial *retention factor*, this congruence between models is credible due to varied geometric approaches each model takes en route to this conclusion: **Model A** uses **maximum likelihood estimation** to find a trend across the entire workforce. **Model C**, however, only cares about the **support vectors** (i.e, the specific employees sitting right on the mathematical boundary between staying and leaving).

In enterprise context this means **Model A** postulates that high anxiety serves as a *retention anchor* for the global workforce while **Model C** offers a more surgical conclusion: for the employees who face imminent attrition risk (the **support vectors** teetering on the edge of the *flight risk* boundary), their level of *risk aversion* (anxiety) is the definitive tie-breaker that keeps them in their seats.

Model E serves to marry some of the divergent perspectives (logistical and psychological) of the models discussed above by **`way_foot (S)`** , **`hr_profession (R)`** , and **`anxiety (M)`** as its top magnitude drivers which lends itself well to the idea that HR managers can seldom afford to overlook the impact of physical logistics (having to walk or take public transit to work) and base line *risk aversion* (i.e employees forming "*continuance committments*" with their employer due to latent or realized fears of job loss).

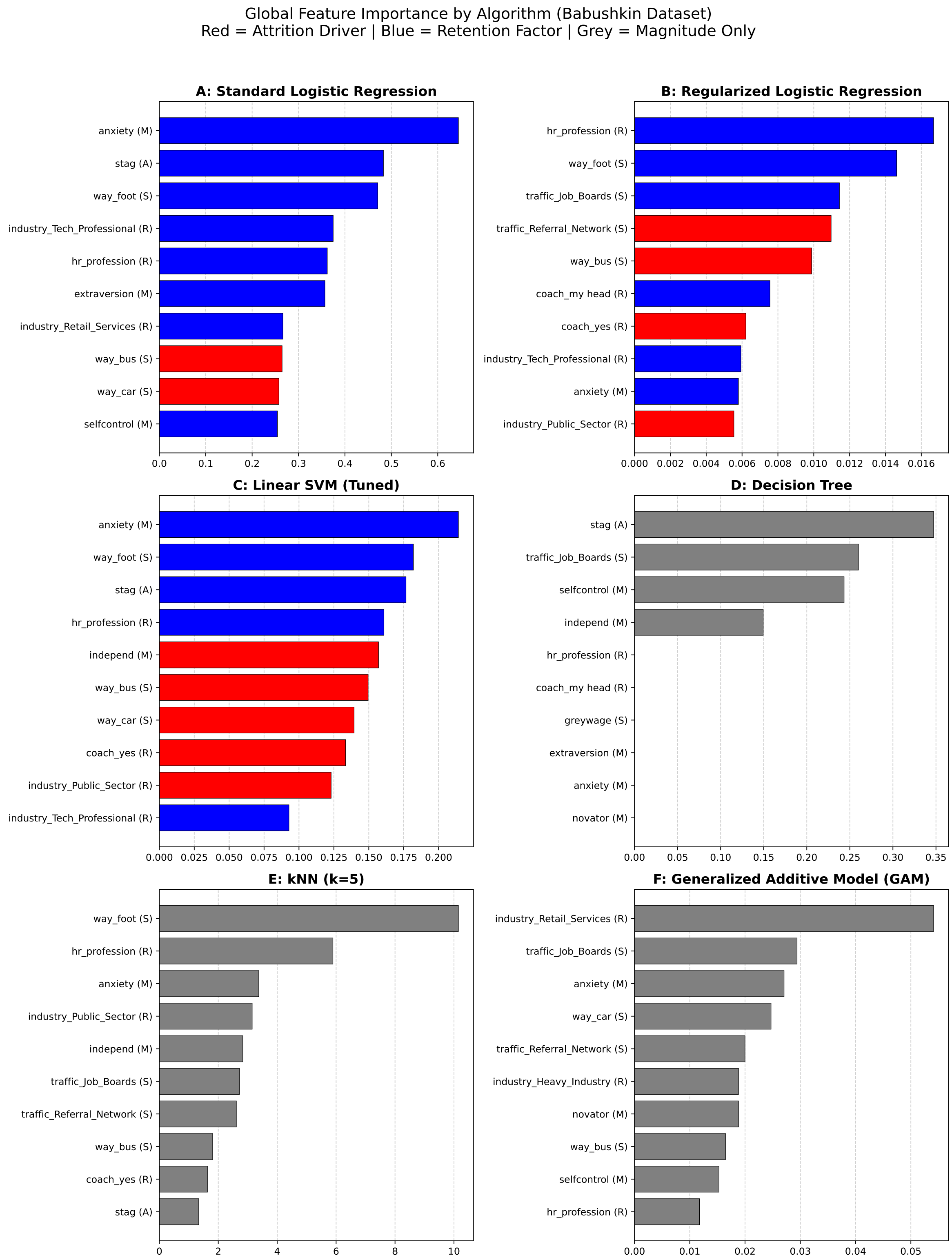


Figure 12: Global Feature Importance by Algorithm (Babushkin Dataset)

Finally, **Model F** violently deviates from any consensus observed in **Models A through E** in its elevation of **`industry_Retail_Services (R)`** and **`traffic_Job_Boards (S)`** to its absolute top spots and in doing so reveals that industry placement (specifically being in Retail) and sourcing origin (Job Boards) possess highly complex, non-linear relationships with turnover at Babushkin. Thereby **Model F** hints that the *attrition curves* for job board sourced retail workers distinguish themselves from their other enterprise counterparts.

3.2.3 Logistic GAM Partial Dependence Plots (Babushkin Dataset)

In this section, the exploration of binary dummy variables via **Partial Dependence Plots** are avoided due to the purpose of **PDP plots** being oriented towards visualizing the marginal effect of a feature across a continuous spectrum. As a result, when **PDP plots** are implemented on binary dummy variables like the **`industry_Retail_Services (R)`** flagged as a top *attrition/retention factor* by **Model F**, the results often reveal themselves to be disjointed step functions with minimal interpretive value. Therefore, in order to sufficiently channel the **spline** capabilities of the **GAM**, the pipeline intentionally bypassed those categorical variables to plot the top continuous drivers: **`anxiety (M)`**, **`novator (M)`**, and **`selfcontrol (M)`**

The top-most visual in Figure 13 which explores the marginal effects of the **`anxiety (M)`** feature points to employee anxiety being a crucial *retention anchor* for the Babushkin enterprise, this is visibly confirmed by observing that for employees with zero anxiety (0.0), the **log odds** of attrition sit dangerously high in the positive *flight risk* territory (~+0.4). However, as the anxiety scale advances to the **X-axis** upper limit of 1.0, the curve plunges cleanly across the **zero-line**, bottoming out at roughly -0.4 and thus furnishes a counter-intuitive behavioral reality in that highly anxious employees are massive *retention anchors*, this also suggests a willingness among *disenamored employees* of the Babushkin organization to retain employment rather than take on a percieved gamble in the wider job market.

The **PDP plot** for **`novator (M)`** is telling even in demonstrating zero **vertical variance** which signals that being an “innovator” has absolutely zero isolated predictive power regarding an employee’s *flight risk* and that initiative and creativity alone cannot translate to *longevity* at the Babushkin organization.

The middle plot in Figure 13 speaks to the virtues of restraint within Babushkin’s workforce. This is because within the **PDP plot** for **`selfcontrol (M)`**, the **log-odds** curve hovers in the slightly positive/neutral zone for employees with low to moderate self-control (0.0 to ~0.5). But once the feature crosses the 0.6 threshold, the **spline** violently dips into the negative *safe zone*. This points to employees with high degrees of self control also having the *self regulative ability* to side step *emotionally driven impulsive decisions* to exit the enterprise.

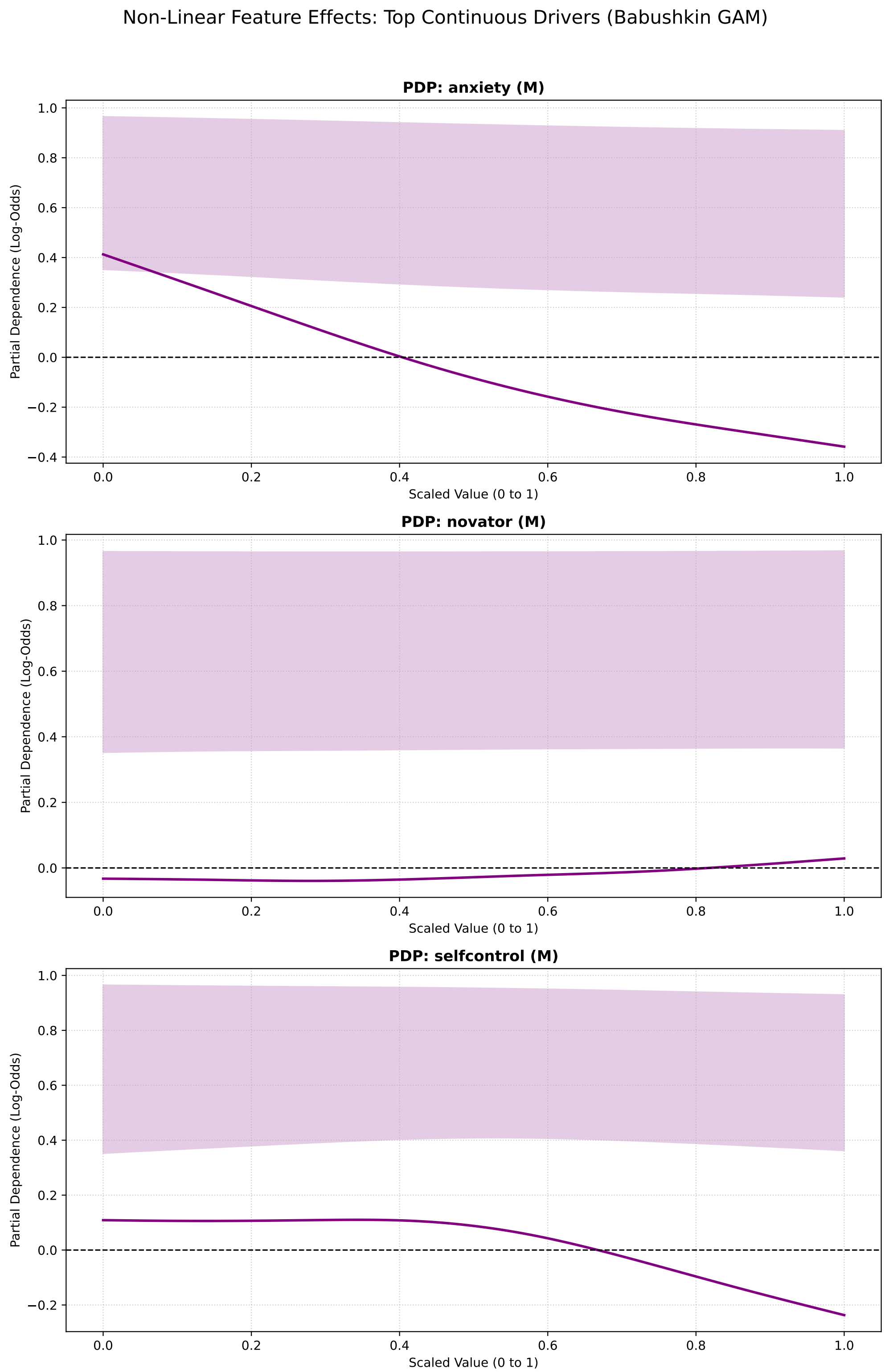


Figure 13: Non-Linear Feature Effects for Top Continuous Drivers (Babushkin GAM)

3.2.4 Decision Tree Attrition "Kill-Zones" (Babushkin Dataset)

```
Kill Zone 1: 63.6% Attrition Concentration
Impact: 0.6/1.0 weighted score (Applies to 261 actual employees | 27.22% of workforce)
Logic Path: stag (A) > 2.73 AND stag (A) <= 27.09 AND independ (M) > 5.15

Kill Zone 2: 55.1% Attrition Concentration
Impact: 0.6/1.0 weighted score (Applies to 256 actual employees | 26.69% of workforce)
Logic Path: stag (A) > 2.73 AND stag (A) > 27.09 AND traffic_Job_Boards (S) <= 0.50

Kill Zone 3: 48.7% Attrition Concentration
Impact: 0.5/1.0 weighted score (Applies to 193 actual employees | 20.13% of workforce)
Logic Path: stag (A) > 2.73 AND stag (A) <= 27.09 AND independ (M) <= 5.15
```

The **Kill zones** above readily reveal the expansive organization wide scope of *turnover* issues at the Babushkin workforce as each of these zones apply to anywhere from 20 to 27 % of the overall work force. This elevated scope suggests that *turnover* at Babushkin may well be more systemic and cultural than initally imagined by the other **predictive models** in past sections.

Kill Zone 1 represents a mid-tenured group (`stag (A)` (*tenure*) between 2.73 and 27.09 months) of self motivated individuals (`independ (M)` $>$ 5.15) and possesses the highest **attrition concentration** at 63.6%, this might serve to guide the Babushkin organization's HR managers in understanding that though employees might've survived *on boarding* and the *learning curve* of their roles,their self motivated natures also empower them to aggressively secure better offers elsewhere if dissatisfied with their employment, this is ostensibly due to the reality that by 27 months of *tenure* in the organizations, most individuals will not have been retained long enough to locked in by deep *seniority* or *institutional inertia*.

Kill Zone 2 explores an identical *tenure group* as sequestered by **Kill Zone 1** but with *job sourcing* being the key differentiator of indicative *turnover intent*, since this group captures over 55% **attrition**, it signals to *HR leaders* that *enterprise talent sourcing strategies* cannot be understated in broader organizational efforts to stem *turnover*.

Kill Zone 3 finds itself as the third rung in a consensus between **decision tree paths** that mid-tenured groups at the Babushkin Organization are particularly beleaguered. This is because all 3 **Kill zones** confront different aspects of the exact same *tenure cohort* (`stag (A)` (*tenure*) between 2.73 and 27.09 months), with **Kill Zone 3** uniquely focusing on employees most in need of *managerial guidance* (`independ (M)` $<$ 5.15).

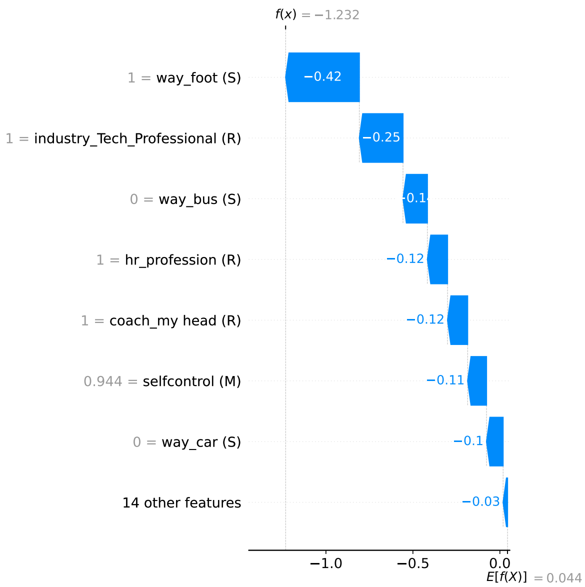
These observations taken altogether in tandem (from **Kill Zones 1 through 3**) inform us that the mid-tenure demographic is the single most vulnerable demographic in the Babushkin organization representing over 50% the workforce and that available remedial steps *enterprise*

HR could take on include paying sufficeint mind to which streams employees are being hired from and the implementation of a dual pronged strategy for the *retention* of mid tenured individuals: for more independent employees (`independ (M)` > 5.15 seen in **Kill Zone 1**) managers must offer extreme *autonomy*, new challenges, and clear *upward mobility* so that *task boredom* does not give way to *professional listlessness* and for more dependent employees (`independ (M)` $<$ 5.15 seen in **Kill Zone 3**), managers must provide heavy *mentorship*, *structural scaffolding*, and *continuous feedback* in order to avoiding burning out or overwhelming these employees.

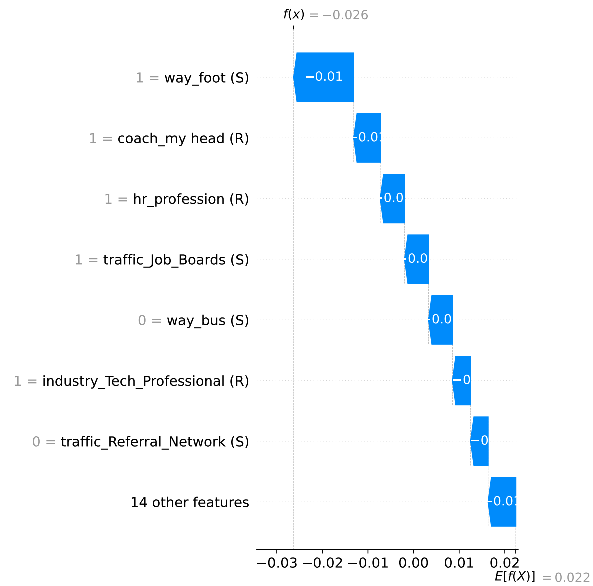
3.2.5 SHAP Extreme Case Analysis: The Safe Bets & *Flight Risks* at Babushkin

Across the diagnostic panel of most *retainable employees* in the Babushkin organization (Figure 14) the **models** paneled here near unanimously agree on the profile of the most stable employees (1 = **`way_foot (S)`** and 1 = **`industry_Tech_Professional (R)`**) by categorizing employees who work in the Tech division and physically walk to work as being most insulated from *attrition risk*. The impact of *direct, active mentorship* from an employee's supervisor is also a crucial driver for *retention* as seen by the prominent placement of the (1 = **`coach_my head (R)`**) variable in Figure 14, this finding is congruent with the canonical belief in HR circles that *adversarial relationships with direct supervisors* predict future *turnover events*.

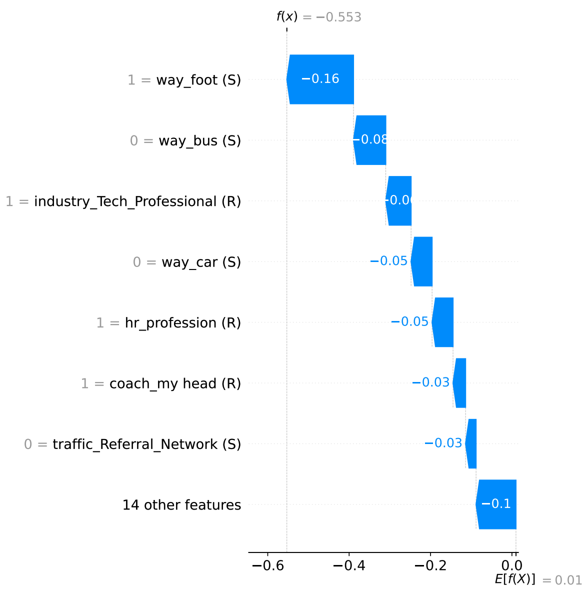
When individualistically assessing the risks at the Babushkin organization via Figure 15, it is apparent that yet another consensus emerges between the **models** about the profiles of the hardest employees to retain. Here the red bars driving the risk are (0.096 = **`anxiety (M)`**) and (0 = **`hr_profession (R)`**), in an HR context, the highest *flight risks* at Babushkin are employees outside the HR department who possess virtually zero *baseline anxiety*. Their fortitude empowers them with the *psychological confidence* to test the *open job market* and their departmental placement outside HR may well also point to a general dissatisfaction with the opacity of *enterprise workforce management decisions*.



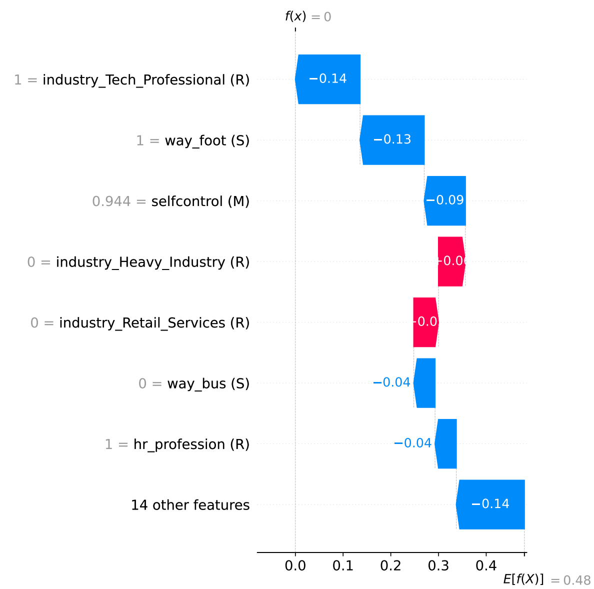
(a) A: Std Logistic Regression - Safest



(b) B: Reg. Logistic Regression - Safest

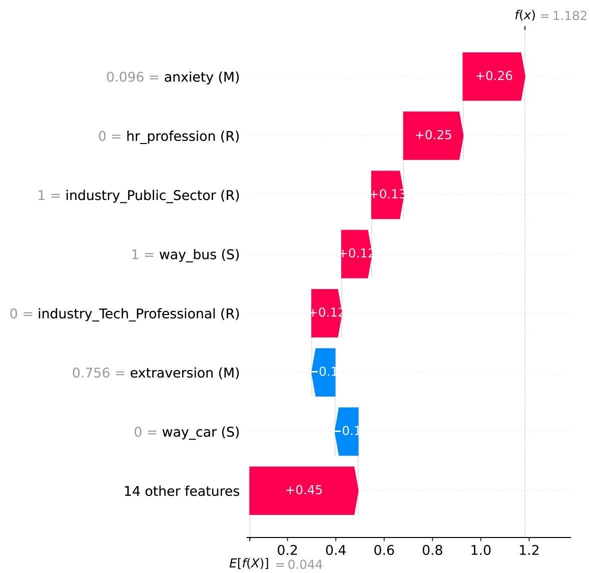


(c) C: Linear SVM (Tuned) - Safest

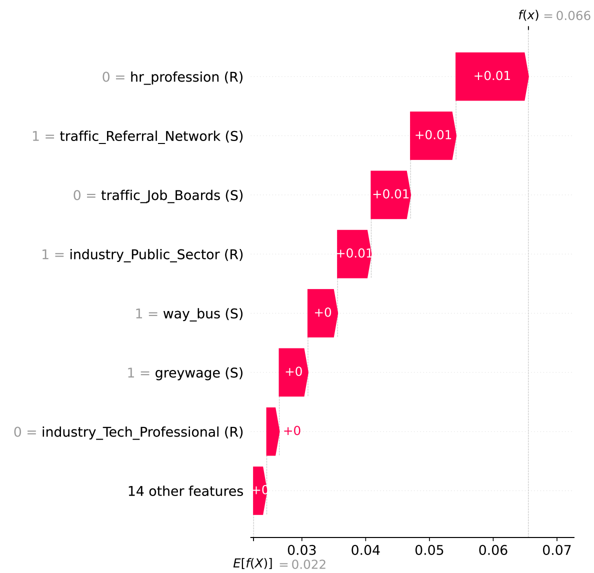


(d) F: GAM - Safest

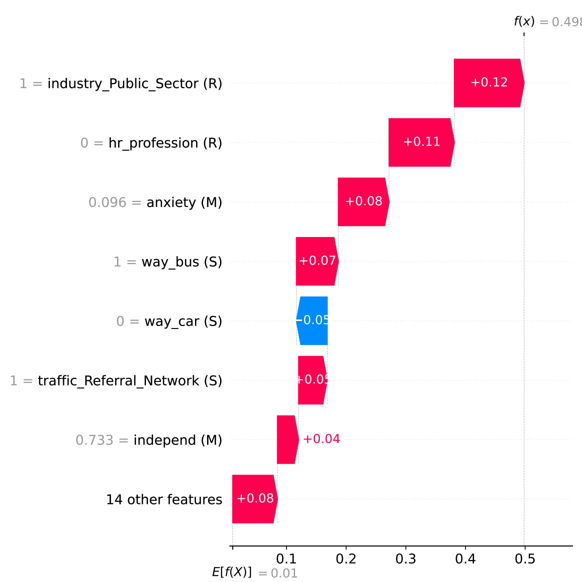
Figure 14: SHAP Local Interpretability: Profiles with Lowest Attrition Probability (Babushkin)



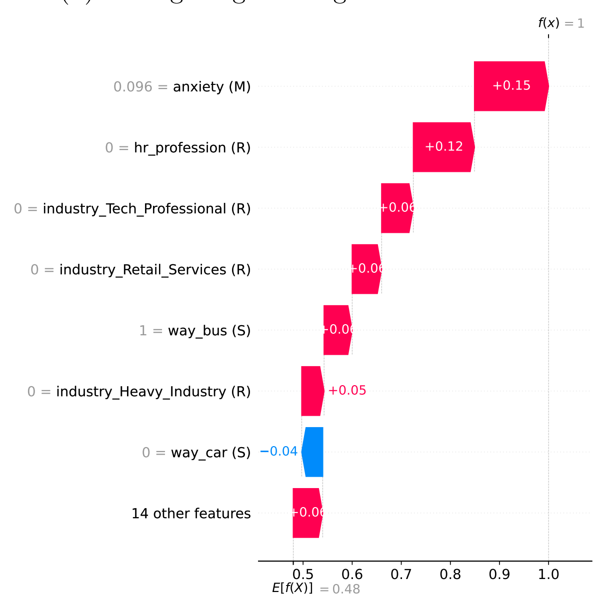
(a) A: Std Logistic Regression - Riskiest



(b) B: Reg. Logistic Regression - Riskiest



(c) C: Linear SVM (Tuned) - Riskiest



(d) F: GAM - Riskiest

Figure 15: SHAP Local Interpretability: Profiles with Highest Attrition Probability (Babushkin)

3.2.6 Model Performance Matrix (Babushkin Dataset)

The **performance matrix** in Table 2 below readily presents itself with a potential pitfall in its highest ranked model (**Model B**). Securing an **Attrition recall** of 0.953 **Model B** seemingly demonstrates a near perfect capability to predict *turnover events*, however as is well understood in the tuning of **turnover classification models** the maximization of **Attrition recall** can only be realized with the sacrifice of **retention recall** as evidenced by **Model B's** retention **recall rate** which stands under 0.1. In practice, the implementation of **Model B** to flag attrition would lead to the squandering of *managerial bandwidth* and resources on *false-positive retention meetings* and related interventions.

This disqualification of **Model B** grounds **Model E (kNN with k = 5)** as the true victor of the contest between **classifier models** in its securing of a highly respectable **Attrition Recall** of 0.721 while maintaining a balanced **Retention Recall** of 0.583. Therefore by utilizing localized and inherently manageable *peer groups* (k=5), **Model E** successfully flags nearly 3 out of 4 actual *flight risks* without the additional risk of profligate *turnover intervention* and thus yields the highest overall **F1-Score** (0.651) of all the tuned **models**.

While an observer of Table 2 may at once notice across the board that F_1 **scores** that top out at 0.6 might be low when held against the **classification metrics** produced by sterile financial models or clinical data, it should be noted that the Babushkin dataset was already perfectly balanced at the onset as mentioned in the introduction of Section 3.2, these **models** were therefore stripped of their ability to artificially inflate their **accuracy** or **F1-Scores** by simply blindly guessing the **majority class** (employees who end up staying). A behavioral explanation for this tendency may lie in the Babushkin dataset's characteristic reliance on psychological traits gathered from self reported survey data which tend to suffer from noisiness and subjectivity as well as bear the risk of potentially dishonest reporting from survey respondents.

Table 2: Consolidated Algorithm Performance and Hyperparameters (Babushkin Dataset)

Rank	model name	tuned hyperparameters	Retention Recall (Class 0)	Attrition Recall (Class 1)	F1-Score (Weighted)
1	B: Regularized Logistic Regression	{'solver': 'saga', 'penalty': 'l2', 'C': 0.001}	0.071	0.953	0.4
2	E: k-Nearest Neighbors	n_neighbors=5	0.583	0.721	0.651
3	F: Generalized Additive Model (GAM)	{'lam': 100.0}	0.56	0.651	0.605

Table 2: Consolidated Algorithm Performance and Hyperparameters (Babushkin Dataset)

Rank	model name	tuned hyperparameters	Retention Recall (Class 0)	Attrition Recall (Class 1)	F1-Score (Weighted)
3	A: Standard Logistic Regression	Baseline (Untuned)	0.56	0.651	0.605
5	C: Linear SVM (Tuned)	{'C': 0.0379269019073225}	0.583	0.616	0.6
6	D: Decision Tree	{'min_samples_split': 6, 'min_samples_leaf': 20, 'max_depth': 3, 'criterion': 'log_loss'}	0.476	0.558	0.517

3.3 Results: Data Science Dataset

As discussed in the Methods section of this study,the Data Science Job Change Dataset is large and imbalanced dataset, therefore the class weights rendered below will be used in the underlying hyperparameter tuning process to robustly penalize classification models for missing the minority class (attrition) in the target variable.

Additionally, in the ETL (extract, transform, load) phase of compution, the columns `experience` and `last_new_job` columns contained string artifacts (like `<1`, `>20`, and `never`) that were coerced into continuous numeric variables by refactoring via Python dictionaries. Furthermore, in the vein of data preservation, missing or null valued categorical data was not dropped and instead imputed with a constant "Unknown" value before one-hot encoding.

```
DS Job Change Dataset Pipeline | Shape Filtered: (19158, 14) -> (19158, 27)
Computed Imbalance Weights: {0: 0.67, 1: 2.01}

Final MARS-Tagged Features:
 - last_new_job (M)
 - experience (A)
 - training_hours (R)
 - city_development_index (S)
 - enrolled_university_Full time course (M)
 - enrolled_university_Part time course (M)
 - enrolled_university_Unknown (M)
 - enrolled_university_no_enrollment (M)
 - relevent_experience_Has relevent experience (A)
 - relevent_experience_No relevent experience (A)
 - company_size_10-100 (R)
 - company_size_100-500 (R)
 - company_size_1000-4999 (R)
 - company_size_10000+ (R)
 - company_size_50-99 (R)
 - company_size_500-999 (R)
 - company_size_5000-9999 (R)
 - company_size_<10 (R)
 - company_size_Unknown (R)
 - company_type_Early Stage Startup (R)
 - company_type_Funded Startup (R)
 - company_type_NGO (R)
 - company_type_Other (R)
 - company_type_Public Sector (R)
 - company_type_Pvt Ltd (R)
 - company_type_Unknown (R)
```

```
did not converge
```

3.3.1 kNN Elbow Diagram & Feature Selection (Data Science Dataset)

As was the case with the IBM and Babushkin datasets in Section 3.1.2 and Section 3.2.1 respectively, Python's Select K-best feature will be used to prune the large feature space down to the most influential 10 variables.

```
kNN will now use only these 10 features:
['last_new_job (M)' 'experience (A)' 'city_development_index (S)'
 'enrolled_university_Full time course (M)'
 'enrolled_university_no_enrollment (M)'
 'relevent_experience_Has relevent experience (A)'
 'relevent_experience_No relevent experience (A)'
 'company_size_Unknown (R)' 'company_type_Pvt Ltd (R)'
 'company_type_Unknown (R)']
```

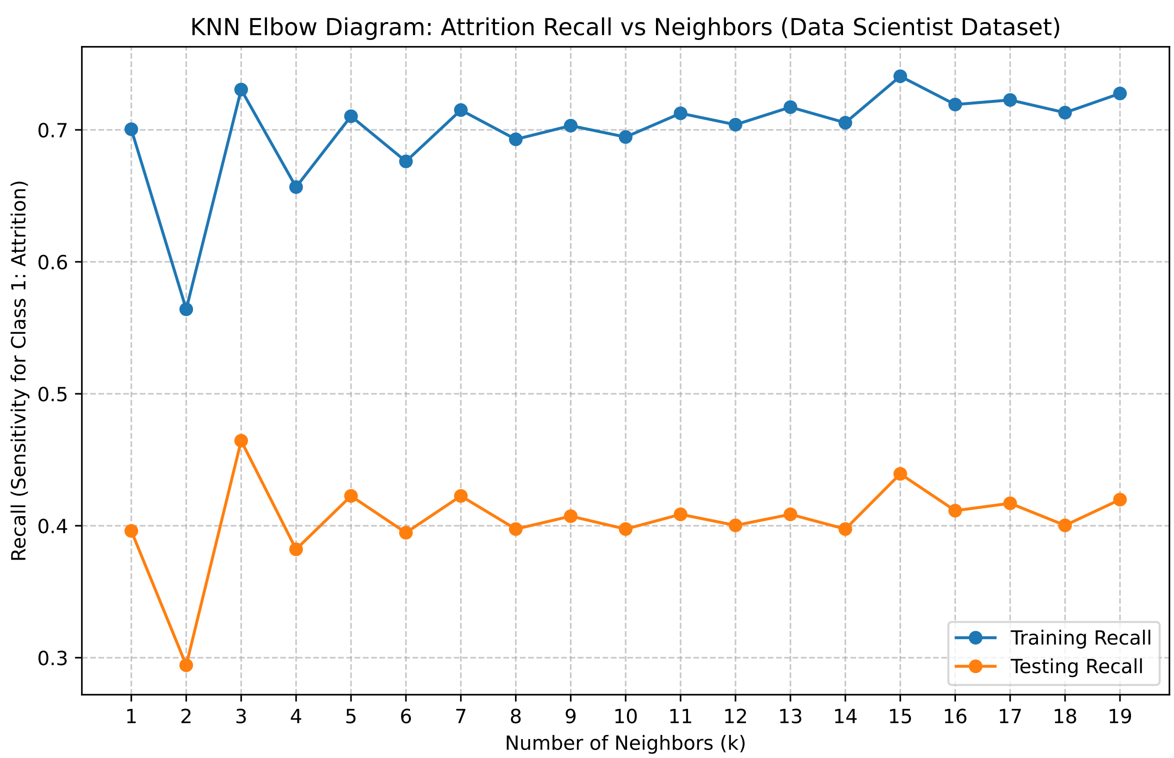


Figure 16: KNN Elbow Diagram: Attrition Recall vs. Neighbors (Data Science Dataset)

```
Peak Performance: k=3 with Test Recall of 0.4644
```

Figure 16 above peaks at $k = 3$, this is consequential in light of the optimal **k-values** produced by the other two datasets in this study (**k=7** for IBM and **k=5** for Babushkin).This hyper-localized k**-value** is likely a direct consequence of the dataset's massive footprint (over 19,000 records) which serves to liberate the **algorithm** from needing to cast a wide geometric net to find statistical peers. It can locate near-exact "clones" within a very tight radius.

The translation of this finding to *Enterprise HR realities* suggest that in massive enterprises (such as Global market leaders and the like), HR need not confine itself to broad *departmental averages* to predict *data scientist turnover*. Large *talent pools* permit the **algorithm** to surgically flag *flight risks* based on the historical behavior of just a hand full of perfectly matched peers, allowing for highly individualized *interventions*.

3.3.2 Global Feature Importances (VIMS Facet Grid - Data Science Dataset)

Figure 17 shows us that the **`city_development_index (S)`** feature achieves something uniquely rare in this study: absolute universal dominance as it is the foremost predictive feature among the entire panel of assembled **models** and in doing so completely overrides the diverging consensus seen in previous sections between **linear and non-linear models**.

In a real word context, while it was observed in Section 3.2 and Section 3.1 that internal *departmental metrics* such as overtime and bad managers drove turnover, *macro-economic gravity* is the lynchpin of *retention* among *Data Science Professionals*. This likely stems from the nature of the industry in sourcing *talent* with highly mobile and premium skillsets and can result in *location hopping* as *Data Science professionals* mature in their careers. This is observable in the panels for **Models A through C** in Figure 17 called out by large **blue bars** which imply that if an emplyee is situated in a highly developed city or tech hub they tend to remain, however if located in a developing or lower tier market they are more likely to translate their experience and acument to roles located in more economically favorable geographies.

Another striking element of Figure 17 is the placement of a synthetically created category (**`company_size_Unknown (R)`**) as the second most powerful **predictive feature** in modeling *attrition*, once again across the entire panel of **predictive models**. This validates the earlier methodological decision to retain **null valued categorical data**. A possible reason for this is when translated to the realities of *workforce dynamics*, missing data is rarely actually random. If a *data scientist* is recorded as working for an enterprise of unknown or unrecorded size, it strongly implies their engagement in a highly volatile *early-stage startup*, a *shell company*, or an unstructured *gig-economy contract* such as free-lancing.

Furthermore, **Models A, B & C** state that having relevant experience is a **predictor** of *attrition* (indicated by **red bars** for the **`relevent_experience_Has relevent experience (A)`** feature). This speaks to the existence of a favorable and lucrative *job market* for mid-level or experienced *data science professionals* and consequently *HR departments* must be cognizant of *poaching risk* as they train and mature their *data science teams*.

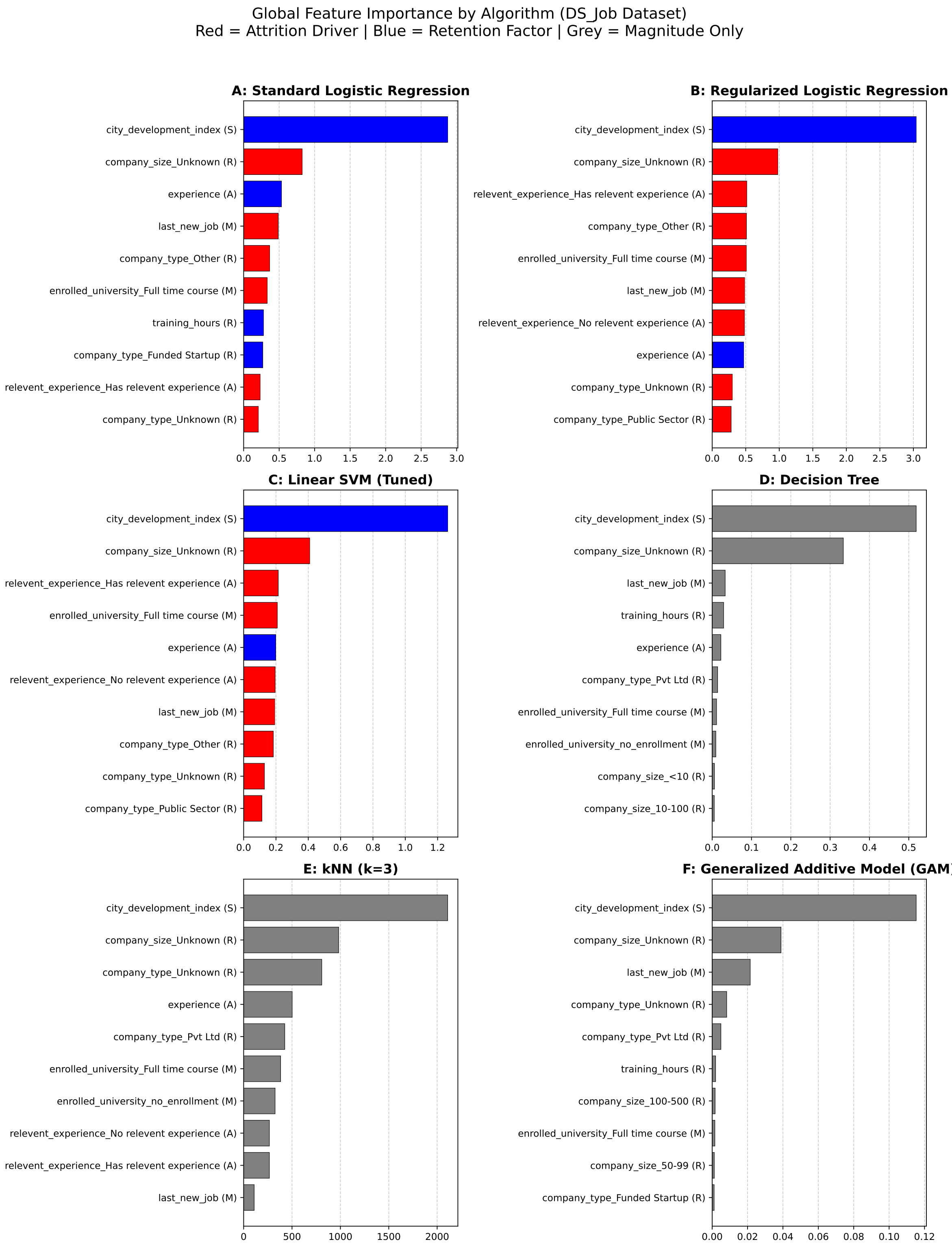


Figure 17: Global Feature Importance by Algorithm (Data Science Dataset)

3.3.3 Logistic GAM Partial Dependence Plots (Data Science Dataset)

Observable at once from Figure 18 is that the **PDP plot** for the **`city_development_index (S)`** feature is that it does not trend downward in a straight line and instead violently spikes of a massive *flight risk* peak at around the 0.3 scale mark, The 0.3 mark can be understood to be *developing or mid-tier economic markets* and *Data Scientists* stationed in these areas pose extreme *flight risk* (well punctuated by the plot peaking at a **log odds** of 2.0 implying an 88% probability of *imminent turnover*). It can be inferred that *Data Scientists* are likely using these roles to embellish their professional resumes before parlaying them into offers at more attractive geographic market(represented by the deep plunge into the negative/*retention zone* at the 0.8 mark and beyond), which instructs us that *enterprise geography* is a revolving door for premium *talent*.

The **PDP plot** for the **`last_new_job`** reveals the potential for *Infrastructural Traps* in the training of *data scientists*. This is visually observable on the plot as risk is deeply negative (safe) at the absolute 0.0 mark but climbs rapidly into the positive *flight-risk zone* and stays there. Green *data scientists* (where **`last_new_job (M) = 0`**) are temporarily anchored because they require massive *organizational infrastructure* such as *cloud computing subscriptions*, *distributed architecture*, and live *databases* to actually mature their skills in contemporary practice. They are thus occupationally and technologically precluded from leaving till they leverage the enterprise to build competencies in *Cloud computing* (like AWS or Azure), once this leverage is secured the *retention anchor* vanishes, and they immediately shop those subsidized skills to competitors.

Similarly, the plot for the **`training_hours`** exposes a 'sweet-spot' for training with an distinguishable nadir (its strongest *retention power*) around the 0.7 scaled mark which it breaks into after elevated *flight risk* seen at lower levels of training. After this localized dip the curve spikes back up toward the 0.9 and 1.0 marks. This indicates that For *data science professionals*, training is not just *corporate onboarding*; it is vital *technological upskilling*. Providing a substantial, heavy investment in their training (the 0.7 mark) secures deep loyalty because the enterprise is actively funding their *human capital*. However this investment poses risk of diminishing returns at the highest levels of training as if developing employees are starved of project work they are consequently starved of *psychological ownership*. The ensuing *task boredom* provides the impetus to direct the company's training resources to certify themselves for a job elsewhere.

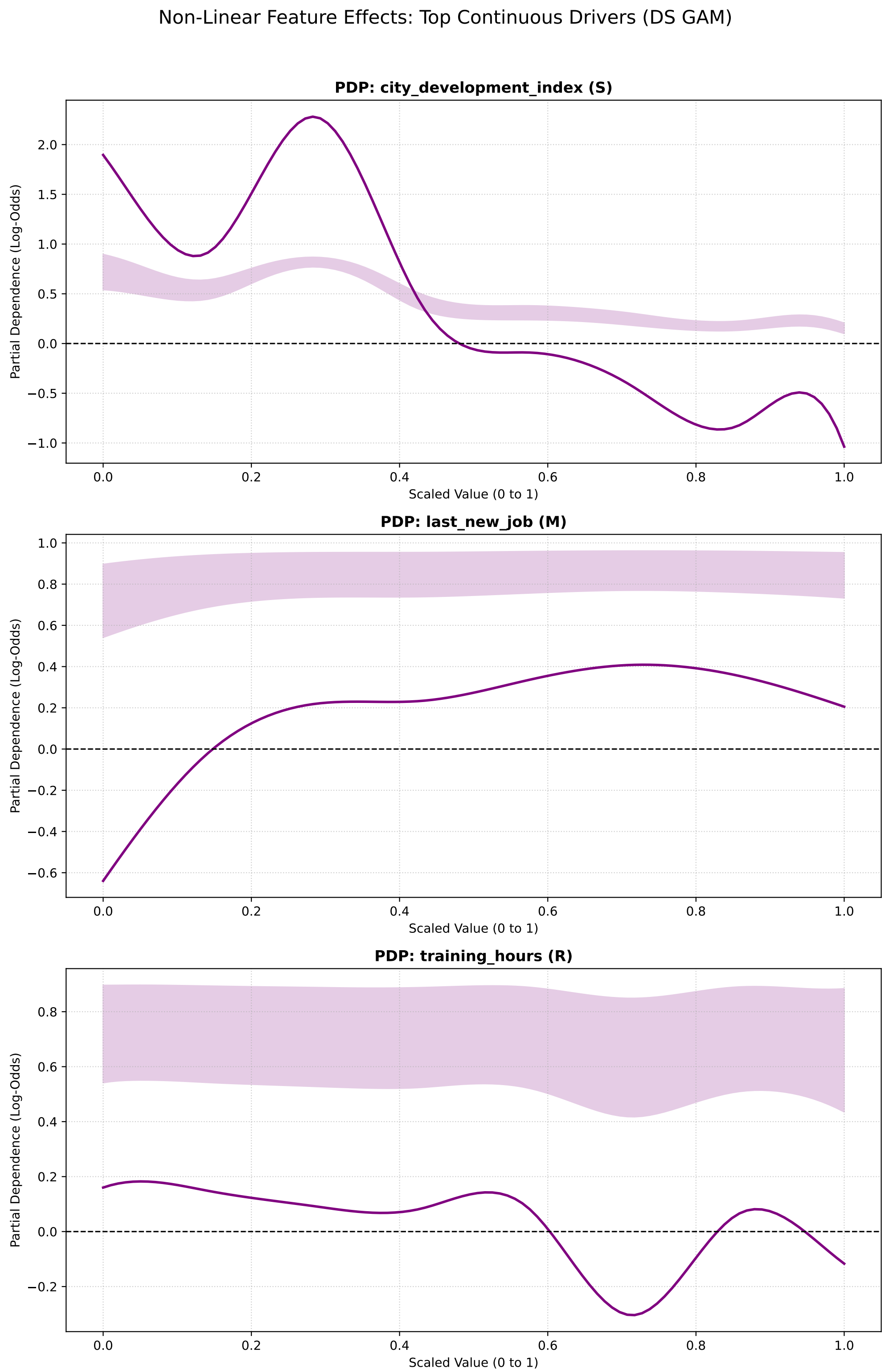


Figure 18: Non-Linear Feature Effects for Top Continuous Drivers (Data Science GAM)

3.3.4 Decision Tree Attrition "Kill-Zones" (Data Science Dataset)

```
Kill Zone 1: 93.8% Attrition Concentration
Impact: 0.9/1.0 weighted score (Applies to 30 actual employees | 0.18% of workforce)
Logic Path: city_development_index (S) <= 0.62 AND experience (A) <= 1.50 AND
relevent_experience_No relevent experience (A) <= 0.50 AND training_hours (R) > 19.50
AND training_hours (R) > 47.50 AND training_hours (R) <= 86.00

Kill Zone 2: 92.9% Attrition Concentration
Impact: 0.9/1.0 weighted score (Applies to 43 actual employees | 0.26% of workforce)
Logic Path: city_development_index (S) <= 0.62 AND experience (A) > 1.50 AND
training_hours (R) <= 73.50 AND company_size_10-100 (R) > 0.50 AND training_hours (R)
> 43.00

Kill Zone 3: 92.6% Attrition Concentration
Impact: 0.9/1.0 weighted score (Applies to 36 actual employees | 0.22% of workforce)
Logic Path: city_development_index (S) <= 0.62 AND experience (A) <= 1.50 AND
relevent_experience_No relevent experience (A) <= 0.50 AND training_hours (R) <=
19.50
```

All three **killzones** in Section 3.3.4 are brought alive by a striking commonality in having the exact same **root node**: **`city_development_index (S) <= 0.62`**. The constructed **decision tree** therfore affirms our earlier inclination from Section 3.3.3 that *Data scientists* stationed in a *developing or lower-tier economic market* are inherently at *risk* thereby grounding the primacy of *geographic reality* in determining *turnover risk* in this industry.

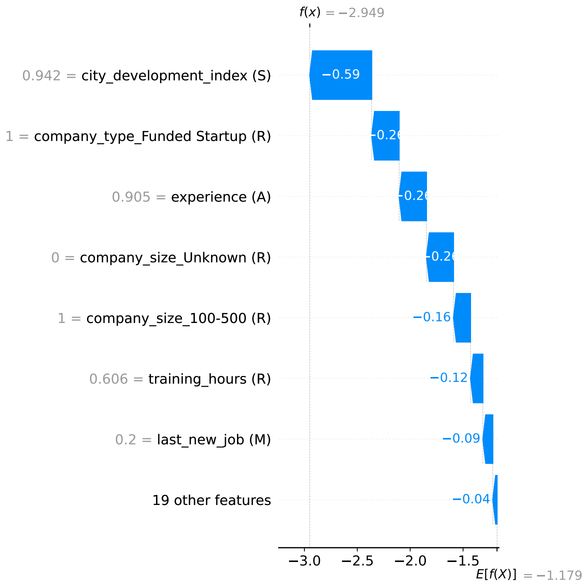
Also revealed in **Kill Zones 1 to 3** are the surgical lethality of *attrition risk* with respect to the *Attrition risk zones* seen in earlier sections like Section 3.2.4 which encapsulated 20-27% of the entire workforce. Here, the Data Science **Kill Zones** are applicale to tiny **micro-cohorts (0.18% to 0.26% of the workforce)**. This indicates that *sectoral turnover* in the Data Science space is not the *company-wide cultural rot* like it was at the Babushkin enterprise but instead localized hyper-specific, and extraordinarily lethal (with **attrition concentrations sitting at a staggering 92.6% to 93.8%**) all but guaraunteeing *turnover* if these specific conditions are met.

We also see divergent groups of *at-risk junior employees* with **Kill Zones 1 and 3**, here both **Kill zones** target a more seasoned cadre *Junior employees* (**`experience` <= 1.50**) who already possess *relevant industry background* (**`relevent_experience_No relevent experience` <= 0.50**) with **training hours** serving to distinguish one cohort from the other as **Kill Zone 1** (**93.8% attrition**) captures juniors receiving moderate/heavy training (**between 47.5 and 86 hours**)and conversely **Kill Zone 3** (**92.6% attrition**) captures juniors receiving almost no training (**`training_hours` <= 19.50**). This speaks to the susceptability of *training programs* in being used as *launch pads* for *voluntary turnover* in the case of **Kill Zone 1** and to *stagnant career prospects* being the *attrition driver* for **Kill zone 3**.

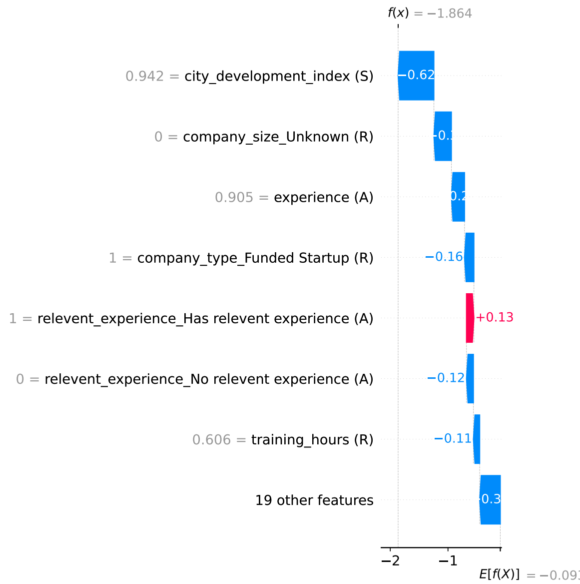
There are noticable glimpses of the *"Big Fish, Small Pond" Syndrome* presented by **Kill Zone 2** (**92.9% attrition**) which targets more *experienced professionals* (**`experience` > 1.50**) in *developing cities* (**`city_development_index` <= 0.62**) working at *small firms* (**`company_size_10-100` > 0.50**) with moderate training (**43 to 73.5 hours**). It is thus apparent that highighly *experienced data scientists* stationed in *small companies* within *mid-tier cities* quickly hit a *professional ceiling*. Sans the massive *computational infrastructure* of a larger enterprise their careers may be destined to outgrow their environment and inevitably defect to *market competitors*.

3.3.5 SHAP Extreme Case Analysis: The Safe Bets & *Flight Risks* for the Data Science dataset

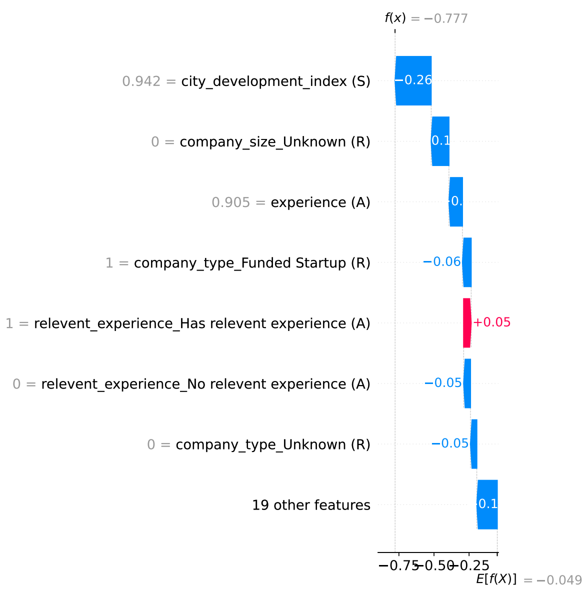
The panel of safest employees produced by Figure 19 validates the *geographical baseline* estabilished in Section 3.3.4 and Section 3.3.3 as the prime driver of *retention* among *Data Science professionals*. As it pertains to *turnover* a noticable contrast also presents itself in these plots between the allure of *agile capital* on the one hand (inunciated by **`1 = company_type_Funded Startup (R)`** as a major **blue** *retention anchor*) and the aversion to *bureaucracy* on the other (seen as **`1 = company_type_Public Sector (R)`** as a **red** *attrition driver* in Figure 20). This speaks to a trend of *data science profesionals* demanding agility backed by capital. They are highly retained by *well-funded startups* where they can build cutting-edge tech but chafe under the pressures endemic to the slow-moving, bureaucratic constraints of the *Public Sector* such as municpal budgets and the reliance of *public sector enterpises* on aging legacy tech stacks.



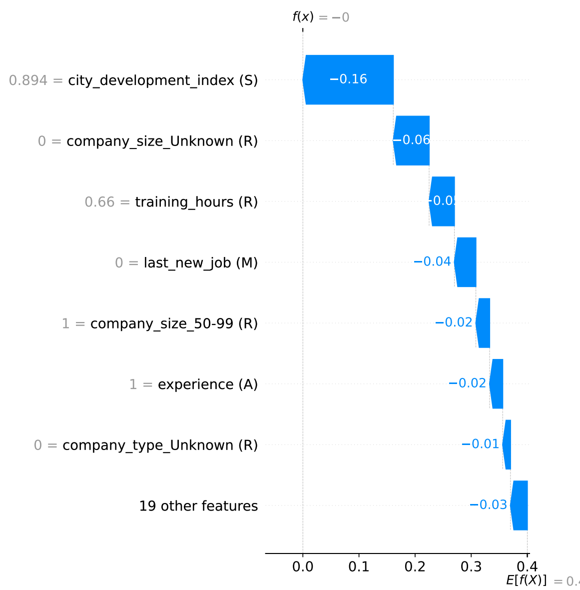
(a) A: Std Logistic Regression - Safest



(b) B: Reg. Logistic Regression - Safest



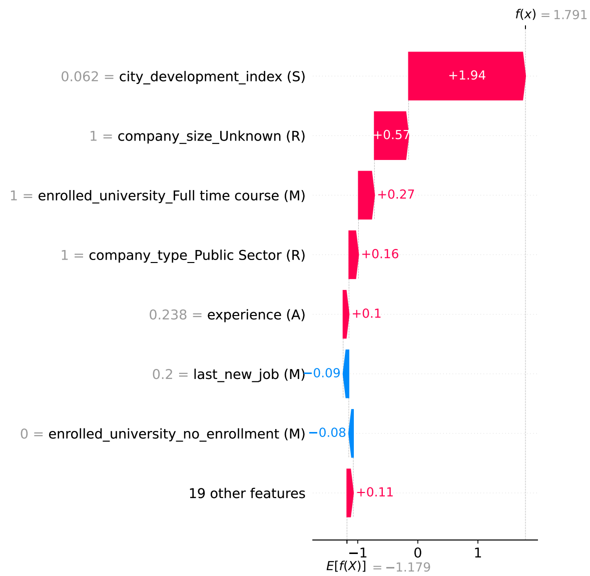
(c) C: Linear SVM (Tuned) - Safest



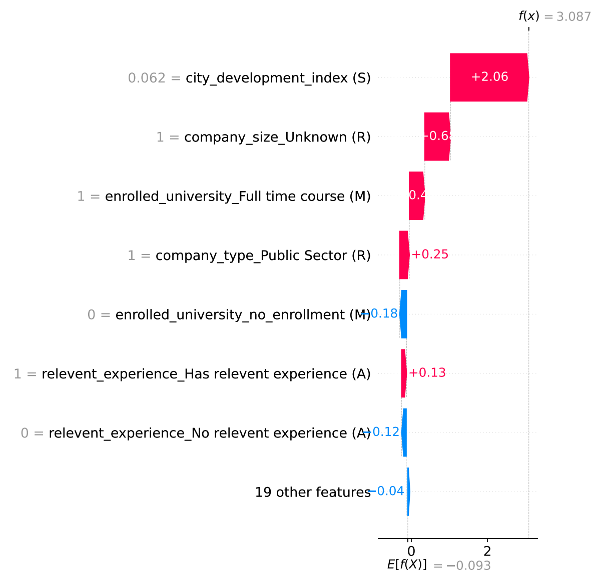
(d) F: GAM - Safest

Figure 19: SHAP Local Interpretability: Profiles with Lowest Attrition Probability (Data Science)

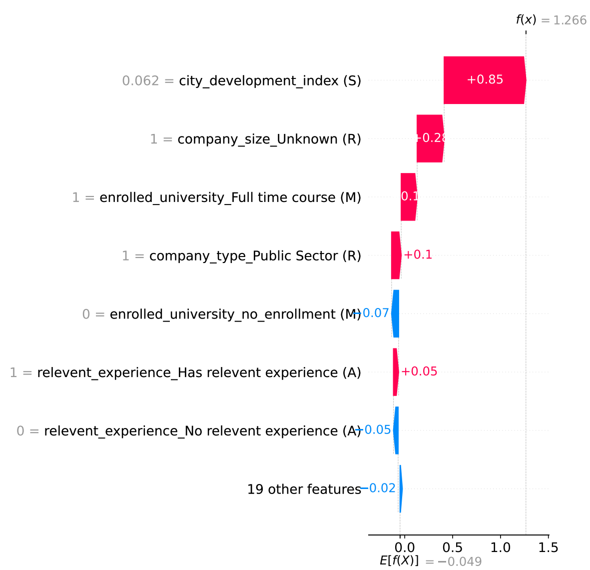
The SHAPs profiles of the most likely leavers rendered in Figure 20 underscores the dangers of *Dual-Track Burnout* as across the **linear models**, **`1 = enrolled_university_Full time course (M)`** is a prominent **red bar** driving the employee's **risk profile** into the *danger zone* which evinces an incapacity among *data scientists* to meet the demands of a *professional career* and a rigorous academic curriculum if both are to take place concurrently. Therefore, *Enterprise HR* must flag full-time students as imminent *flight risks* as the cognitive load of this highly specialized, computationally heavy profession leaves no bandwidth for a full academic course load.



(a) A: Std Logistic Regression - Riskiest



(b) B: Reg. Logistic Regression - Riskiest



(c) C: Linear SVM (Tuned) - Riskiest

(d) F: GAM - Riskiest

Figure 20: SHAP Local Interpretability: Profiles with Highest Attrition Probability (Data Science)

3.3.6 Model Performance Matrix (Data Science Dataset)

Table 3: Consolidated Algorithm Performance and Hyperparameters (Data Science Dataset)

Rank	model name	tuned hyperparameters	Retention Recall (Class 0)	Attrition Recall (Class 1)	F1-Score (Weighted)
1	D: Decision Tree	{'min_samples_split': 7, 'min_samples_leaf': 30, 'max_depth': 7, 'criterion': 'log_loss'}	0.737	0.788	0.765
2	F: Generalized Additive Model (GAM)	{'lam': 0.1}	0.745	0.778	0.768
3	B: Regularized Logistic Regression	{'solver': 'saga', 'penalty': 'l2', 'C': 444.445}	0.694	0.777	0.733
4	C: Linear SVM (Tuned)	{'C': 0.0379269019073225}	0.7	0.77	0.735
5	E: k-Nearest Neighbors	n_neighbors=3	0.84	0.464	0.744
6	A: Standard Logistic Regression	Baseline (Untuned)	0.937	0.264	0.735

Table 3 demonstrates the unsuitability of standard **untuned and unoptimized models** on complex and imbalanced datasets via **Model A**'s dismal **Attrition recall** of **0.264** which implies the **model** fails in identifying 3 out of 4 employees at risk of *attrition* which affirms heavily the **ensemble approach** undertaken by this study.

A similar deficiency was recorded in the performance of **Model E (kNN algorithm)** which despite offerring fascinating conceptual insights into *hyper-localized peer groups* ultimately floundered in **global predictive power** by securing an **Attrition Recall** of only **0.464**. This shows that while concluding that *turnover* may be inferred by a *data scientist's* three closest peers may be intuitive and digestible to *HR practicioners* it is not robust enough to serve as an *enterprise wide early warnings system* as it pertains to this dataset which is further substantiated by the observed outperformance by hard the bifurcations of the **Decision Tree** and the splines of the **GAM**.

The aforementioned **Decision tree (Model D)** emerges as the victor of the contest between **predictive models** by achieving the highest **Attrition Recall** in the panel at **0.788** while also maintaining a highly respectable **Retention Recall** of **0.737**. This means the **decision tree** flags almost 80 percent of *flight risk* without overcommitting *HR* to specious *retention meetings* and also adds to the validity of the *attrition kill zones* conceptulized in Section 3.3.4.

Model D is not alone in this efficacy and is closely followed by **Model F (Logistic GAM)** in performance and secures the highest F_1 **score** of the **ensemble** of **0.768** in tow with a **recall** only **1%** lower than **Model D**'s at **0.778**. This performance equips *enterprise HR* with a more conservative alternative than **Decision trees** if so desired by offering a near identical computational and predictive safety net.

4 Limitations and Confounds of Study

The Pipeline conceptualized and built in this study is subject to a following limitations and confounds:

- Synthetic Baseline Confound (IBM Dataset): Because this data set was fashioned by *Data Scientists* and wasn't compiled from documented or recorded human behavior it produces an unnatural mathematical cleanliness that cannot be replicated in *enterprise HR practice* owing to the irrational, chaotic "noise" of *real-world workforce dynamics* which often leads to messy data replete with **missing values** and incorrectly entered data.
- Cultural Translation Pitfalls(Babushkin Dataset): Though not synthetic (like the IBM dataset) as it is an extraction from an Russian HRIS system, there yet exists a deep structural sparsity in this dataset, the cultural confound identified here is most saliently observed with variables like `greywage` that document endemic economic conditions not present globally. By coercing these localized variables into universal buckets like *"Situational Factors"*, the pipeline risks misinterpreting culturally specific *survival tactics* as standard *corporate behaviors* which could result in misdirection if adopted by multinational or foreign enterprises.
- Adminstrative Masking of Information Gaps (Data Science Dataset): As introduced in Section 3.3, **missing or null valued data** was subjected to a **label imputation strategy** where every instance of **null values** in **categorical variables** was replaced with the **"Unknown"**. While this *data conservation practice* enabled the garnering of insights such as `company_size_Unknown` being discovered as a massive *flight risk signal* this **imputation strategy** is subject to the risk of **misclassifaction**. This is because treating the absence of data as a definitive behavioral feature assumes that the data is missing on purpose and dismisses the rather significant possibility of *organizational or departmental indolence*, *human error* and *data corruption* being the causes of missing categorical information.

- The Static Illusion of Cross Sectional Data (Global Constraint): As all 3 datasets operationalzied by this study provide *cross sectional snapshots* rather than *longitudinal workforce assessments*. This could lead to costly oversimplifcations as human behavior and *motivation* are not static. If algorithmn are only able to adapt to snapshot data they cannot account for the trajectory of features over time such as *morale*. Using the IBM dataset as an example, if an employee had a **`JobSatisfaction`** score of 2 (on a scale of 0 to 4), this would suggest to the reader that the employee is a *flight risk* if the variable is interpretted under temporal isolation, however had there been additional context that the same employee's **`JobSatisfaction`** from a prior period was a score of 1, their score of 2 in the current period would instead divulge the incremental success of *retention efforts*.

5 Conclusion

The defining characteristic of this study was the integration of the **MARS Model** (an *Organizational Behavior Framework*) with **Attrition Classification analysis** and subsequently the preferential integration of interpretable **Classification models** over their **black box** counterparts. The former aim which married the principles of *data science* to those of *data management*, allowed for the diagnosis of complex *human centric phenomena* like *Dual-Track Burnout* (Section 3.3.5), *Infrastructural Traps*(Section 3.3.3), and the dangers of *"greywages"* (Section 4) which would have been imperceptible under the auspices of standard **numerical analysis**. The latter aim served to orient the output of this study in a praxis forward manner to the realities of *enterprise HR*. While foundational literature often presumes a strict trade-off between predictive accuracy and interpretability, the empirical reality of the Babushkin dataset challenges that dichotomy. Contemporary community benchmarks reveal that opaque **ensemble methods** like **Random Forests** typically plateau at **F1 scores** between 0.68 and 0.80 on this specific dataset. Given that the interpretable **models** in this study generated **F1 scores** spanning 0.400 to 0.857, they achieved highly competitive *predictive parity* without sacrificing structural transparency. Simply put, **blackboxes** might match our efficacy in informing us "who" will leave but remain mathematically incapable of informing us why. Hence the adoption of interpretable **models** enables **algorithmic logic** to better conform to business needs by providing transparent and operational logic rather than opaque prophecies.

Furthermore, **Decision Trees (Model D)** and **Logistic GAM's (Model F)** emerged from the **modelling ensemble** as the most viable *enterprise tools* of the lot. In their isolation of otherwise subterranean *flight risks* (via **Decision Tree** *"Kill Zones"*) and their mapping of behavioral inflection points (via **GAM PDP** *"Sweet spots"*) they affirm that *HR departments* can proactively target *attrition* without relinquishing their data to unexplainable **algorithms**.

The future scope for the enhancement of this methodology can be undertaken from multiple paths. One path would lie in transcending the use of cross sectional data operationalzied throhout this study via the compilation and standardization of longitudinal or time series datasets which secures the ability to track the *"temporal velocity"* of an employee's *morale* and *performance* over multiple quarters, rather than judging them on a single, static day.

A second more action oriented path would involve the embedding of *"Kill Zone"* filters in a live *corporate environment* to facilitate pilot assessments of these filters in observing if *HR maqnagers* utilizing **MARS** insights might materially reverse adverse *turnover predictions* in real time with *interventional strategy.*

While featuring an **ensemble of models** grounded in different **mathematical perspectives**, other interpretable **models** such as **K-Means Clustering** and **Stochastic Gradient Descent classifiers** may be paneled in future studies to ameliorate computational diversity. Similarly, while the use of techniques like the **constant imputation strategy** utilized in this study produces misinterpretive risk (Section 3.3), future iterations of this pipeline may benefit from the inclusion of advanced **imputation techniques** (like **MICE - Multiple Imputation by Chained Equations**). The diagnostic capabilites of this pipeline may also be bolstered with **Individual Conditional Expectation (ICE) plots** which visualize how a **machine learning model's predictions** change for each observation when a specific feature varies, offers an alternative or a complementary inclusion to **SHAPs plots** for localized and individual inspections of *attrition risk.*

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